

Supplementary Materials for “High Dimensional Gaussian Graphical Regression Models with Covariates”

S1 Additional Numerical Results

S1.1 Sensitivity analysis

We conduct a sensitivity analysis to examine the performance of our method when σ^{jj} is covariate dependent. Specifically, under the same setting as in Section 6, we set $\sigma^{jj}(\mathbf{u}) = 1 + \sum_{h=1}^q \beta_{jjh} u_h$. If u_h is an effective covariate with a nonzero effect on the graph, we set s_σ proportion of $\{\beta_{jjh}\}_{j \in [p]}$ to σ_0 , where σ_0 reflects the strength of covariate dependence for $\sigma^{jj}(\mathbf{u})$, and set the rest of β_{jjh} ’s to zero. We let $n = 200$, $p = 25$ and $s_\sigma = 0.1$. The tuning parameters are selected using cross validation. Table S1 reports the average evaluation criteria over 200 data replicates over $q = 25, 100$ and $\sigma_0 = 0, 0.1, 0.2$. Note that $\sigma_0 = 0$ corresponds to the case that σ^{jj} ’s are not covariate dependent. When $\sigma_0 = 0.2$, we have $\sigma^{jj}(\mathbf{u}) \in [1, 2]$. Reported criteria are estimation errors of the mean $\boldsymbol{\mu}_{\text{error}} = \sum_{i=1}^n \|\hat{\boldsymbol{\mu}}_i - \boldsymbol{\mu}_i\|^2/n$ and the precision matrix $\boldsymbol{\Omega}_{\text{error}} = \sum_{i=1}^n \|\hat{\boldsymbol{\Omega}}_i - \boldsymbol{\Omega}_i\|_{F,\text{off}}^2/n$, where $\|\cdot\|_{F,\text{off}}$ denotes the off-diagonal Frobenius norm and $(\boldsymbol{\mu}_i, \boldsymbol{\Omega}_i)$ and $(\hat{\boldsymbol{\mu}}_i, \hat{\boldsymbol{\Omega}}_i)$ are the true and estimated values, respectively, the true positive rate (TPR) and false positive rate (FPR) in estimating the precision coefficient $(\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_p)$. As seen from Table S1, under this misspecified setting with covariate dependent residual variances, our method still gives a reasonable performance.

S1.2 Estimation accuracy of $\boldsymbol{\Gamma}$

Table S2 shows that the first step of our estimation procedure achieves good performance, and the estimation error of $\boldsymbol{\Gamma}$ decreases as n increases, or as p and q decrease. Such observations conform to Theorem 3.

		μ_{error}	Ω_{error}	TPR_{β}	FPR_{β}
$q = 50$	$\sigma_0 = 0$	3.041 (0.018)	2.011 (0.018)	0.817 (0.004)	0.003 (0.000)
	$\sigma_0 = 0.1$	2.871 (0.022)	2.167 (0.015)	0.700 (0.004)	0.004 (0.000)
	$\sigma_0 = 0.2$	2.842 (0.018)	2.663 (0.020)	0.645 (0.004)	0.003 (0.000)
$q = 100$	$\sigma_0 = 0$	3.603 (0.019)	2.147 (0.016)	0.777 (0.005)	0.002 (0.000)
	$\sigma_0 = 0.1$	3.465 (0.022)	2.382 (0.016)	0.668 (0.004)	0.002 (0.000)
	$\sigma_0 = 0.2$	3.370 (0.023)	2.878 (0.018)	0.627 (0.003)	0.002 (0.000)

Table S1: Average evaluation criteria with varying covariate dimension q and covariate dependence parameter σ_0 with standard errors in parenthesis.

n	(p, q)	TPR_{Γ}	FPR_{Γ}	Error of Γ
200	(25,50)	0.872 (0.003)	0.156 (0.002)	1.811 (0.006)
	(25,100)	0.804 (0.003)	0.095 (0.001)	1.980 (0.006)
	(50,50)	0.867 (0.003)	0.110 (0.001)	1.938 (0.006)
	(50,100)	0.789 (0.003)	0.072 (0.000)	2.162 (0.005)
400	(25,50)	0.996 (0.000)	0.172 (0.001)	1.273 (0.004)
	(25,100)	0.986 (0.001)	0.113 (0.001)	1.386 (0.005)
	(50,50)	0.992 (0.001)	0.122 (0.000)	1.351 (0.004)
	(50,100)	0.987 (0.001)	0.082 (0.001)	1.514 (0.004)

Table S2: Estimation accuracy of Γ in simulations with various sample sizes n , network sizes p and covariate dimensions q .

S1.3 Higher dimensional cases

We have furthered increased (p, q) to $(25, 400)$ and $(300, 300)$. The simulation results over 200 data replicates are reported in Table S3. For both settings in Table S3, the mean coefficient Γ are set following Section 6. The non-sparse entries in $\{\mathbf{B}_h\}_{h=0}^q$ for $(p, q) = (25, 400)$ and $(p, q) = (300, 300)$ are set to be the same as $p = 25$ and $p = 50$ in Section 6, respectively. These settings involve an extremely large number of parameters; for example, with $(p, q) =$

(300, 300), the total number of precision parameters is almost 27 million (or 26,999,700 precisely). It appears that our proposed method still achieves a reasonable performance in these high dimensional settings.

n	(p, q)	Error of $\mathbf{\Gamma}$	Error of $\mathbf{\beta}$	TPR $_{\mathbf{\beta}}$	FPR $_{\mathbf{\beta}}$
200	(25, 400)	2.195 (0.005)	1.534 (0.005)	0.721 (0.004)	0.0003 (0.0000)
	(300, 300)	2.525 (0.007)	2.461 (0.005)	0.527 (0.004)	0.0000 (0.0000)

Table S3: Estimation and selection accuracy of **RegGMM** with larger network sizes p and covariate dimensions q .

S1.4 Comparison with benchmark methods

We compare our proposed method with several benchmark solutions, including the standard i.i.d Gaussian graphical model estimated by the neighborhood selection method ([Meinshausen and Bühlmann, 2006](#)), referred to as **MB**, and by the graphical lasso estimation method ([Friedman et al., 2008](#)), referred to as **glasso**, the conditional mean Gaussian graphical model ([Cai et al., 2012](#)), referred to as **cmGMM**, and the stratified Gaussian graphical model ([Danaher et al., 2014](#)), referred to as **stGMM**. Our evaluation criteria include the average estimation error of the mean defined as $\boldsymbol{\mu}_{\text{error}} = \sum_{i=1}^n \|\hat{\boldsymbol{\mu}}_i - \boldsymbol{\mu}_i\|^2/n$, where $\boldsymbol{\mu}_i = \mathbf{\Gamma} \mathbf{u}^{(i)}$ and $\hat{\boldsymbol{\mu}}_i$ is estimated from a given method, average estimation error of the precision matrix defined as $\boldsymbol{\Omega}_{\text{error}} = \sum_{i=1}^n \|\hat{\boldsymbol{\Omega}}_i - \boldsymbol{\Omega}_i\|_{F, \text{off}}^2/n$, where $\boldsymbol{\Omega}_i = \mathbf{B}_0 + \sum_{h=1}^q \mathbf{B}_h \mathbf{u}_h^{(i)}$ and $\hat{\boldsymbol{\Omega}}_i$ is estimated from a given method, and the average selection error of the precision matrix $\text{TPR}_{\boldsymbol{\Omega}} = \sum_{i=1}^n \text{TPR}(\hat{\boldsymbol{\Omega}}_i)/n$ and $\text{FPR}_{\boldsymbol{\Omega}} = \sum_{i=1}^n \text{FPR}(\hat{\boldsymbol{\Omega}}_i)/n$, where $\text{TPR}(\hat{\boldsymbol{\Omega}}_i)$ and $\text{FPR}(\hat{\boldsymbol{\Omega}}_i)$ are true and false positive errors calculated by comparing the entries in $\hat{\boldsymbol{\Omega}}_i$ and $\boldsymbol{\Omega}_i$, respectively.

With both **MB** and **glasso** stemming from an i.i.d. model, we have $\hat{\boldsymbol{\mu}}_i = \sum_{j=1}^n \mathbf{x}^{(j)}/n$ and $\hat{\boldsymbol{\Omega}}_i = \hat{\boldsymbol{\Omega}}$, where $\hat{\boldsymbol{\Omega}}$ is the output from **MB** or **glasso**. As **cmGMM** assumes heterogeneous

n	p, q	Method	μ_{error}	Ω_{error}	TPR_{Ω}	FPR_{Ω}
200	$p = 25$ $q = 50$	RegGMM	3.042 (0.019)	2.011 (0.018)	0.853 (0.004)	0.122 (0.002)
		MB	7.067 (0.010)	3.371 (0.011)	0.666 (0.003)	0.129 (0.002)
		glasso	7.067 (0.010)	10.824 (0.027)	0.706 (0.004)	0.158 (0.002)
		cmGMM	3.919 (0.040)	29.248 (0.443)	0.982 (0.003)	0.477 (0.002)
		stGMM	7.219 (0.015)	8.807 (0.054)	0.492 (0.008)	0.003 (0.001)
	$p = 25$ $q = 100$	RegGMM	3.603 (0.019)	2.147 (0.016)	0.809 (0.005)	0.121 (0.002)
		MB	6.916 (0.010)	3.168 (0.012)	0.642 (0.004)	0.107 (0.002)
		glasso	6.916 (0.010)	11.394 (0.040)	0.698 (0.004)	0.149 (0.002)
		cmGMM	5.724 (0.023)	15.236 (0.225)	0.837 (0.005)	0.340 (0.004)
		stGMM	7.226 (0.011)	9.131 (0.047)	0.502 (0.007)	0.028 (0.001)
	$p = 50$ $q = 50$	RegGMM	3.753 (0.020)	5.036 (0.024)	0.657 (0.004)	0.104 (0.002)
		MB	7.735 (0.010)	6.650 (0.014)	0.448 (0.003)	0.060 (0.001)
		glasso	7.735 (0.010)	15.661 (0.031)	0.511 (0.002)	0.090 (0.001)
		cmGMM	5.491 (0.027)	34.644 (0.232)	0.890 (0.003)	0.430 (0.002)
		stGMM	8.414 (0.013)	11.345 (0.071)	0.182 (0.008)	0.003 (0.000)
	$p = 50$ $q = 100$	RegGMM	4.440 (0.019)	5.332 (0.020)	0.629 (0.003)	0.094 (0.001)
		MB	7.791 (0.010)	6.656 (0.015)	0.447 (0.002)	0.061 (0.001)
		glasso	7.791 (0.010)	15.697 (0.039)	0.524 (0.003)	0.095 (0.001)
		cmGMM	6.199 (0.028)	29.146 (0.080)	0.829 (0.003)	0.396 (0.000)
		stGMM	8.667 (0.013)	11.322 (0.077)	0.182 (0.008)	0.004 (0.000)

Table S4: Evaluation criteria with varying network size p and covariate dimension q . Numbers achieving the best evaluation criteria in each setting are marked in boldface.

means but a common covariance, we let $\hat{\mu}_i = \hat{\Gamma} \mathbf{u}^{(i)}$ and $\hat{\Omega}_i = \hat{\Omega}$, where $\hat{\Gamma}$ and $\hat{\Omega}$ are the output from cmGMM. For stGMM and given q binary covariates, there are 2^q groups (or conditions) to be considered, far exceeding the number of subjects n . To implement stGMM, we cluster the subjects into $K < n$ groups by applying K -means clustering to the covariates and denote the cluster label of subject i as $c_i \in \{1, \dots, K\}$. We then implement stGMM and let $\hat{\mu}_i = \sum_{j=1}^n \mathbf{x}^{(j)} \mathbf{1}_{\{c_j=c_i\}} / (\sum_{j=1}^n \mathbf{1}_{\{c_j=c_i\}})$ and $\hat{\Omega}_i = \hat{\Omega}_{c_i}$, where $\Omega_1, \dots, \Omega_K$ are the output from stGMM. All benchmark methods are tuned using the tuning methods suggested in the respective papers.

Under the same setting as in Section 6 and let $p = 25, 50$, $q = 50, 100$, $n = 200$ and $K = 5$, Table S4 reports the average evaluation criteria over 200 data replicates, revealing that **RegGMM** improves upon the benchmark methods notably in estimation accuracy and selection accuracy. Specifically, because **MB**, **glasso** and **stGMM** are not designed to estimate the covariate dependent means or precision matrices well, they tend to incur large estimation errors and low selection accuracy. While **cmGMM** estimates well the covariate dependent means, it is not designed to accommodate covariate dependent precision matrices and, therefore, tends to overselect nonzero coefficients in $\hat{\Omega}$ in the settings examined.

S1.5 Role of standardization

We comment on the model interpretability after standardizing covariates (Tibshirani, 1997) to have mean zero and variance one. As such, each graphical regression (8) is in effect

$$\hat{z}_j = \sum_{k \neq j}^p \beta_{jk0} \frac{\hat{z}_k}{\text{sd}(z_k)} + \sum_{k \neq j}^p \sum_{h=1}^q \beta_{jkh} \frac{(\mathbf{u}_h - \bar{\mathbf{u}}_h)}{\text{sd}(\mathbf{u}_h)} \odot \frac{\hat{z}_k}{\text{sd}(\hat{z}_k)} + \epsilon_j,$$

where $\bar{\mathbf{u}}_h$ denotes the mean of $\mathbf{u}_h$, $\text{sd}(\cdot)$ denotes the standard deviation, and $\hat{z}_j$, residual from the first estimation step, has mean zero. The above equation can be re-written as

$$\hat{z}_j = \sum_{k \neq j}^p \tilde{\beta}_{jk0} \hat{z}_k + \sum_{k \neq j}^p \sum_{h=1}^q \tilde{\beta}_{jkh} \mathbf{u}_h \odot \hat{z}_k + \epsilon_j, \quad (\text{S1})$$

where $\tilde{\beta}_{jk0} = \frac{1}{\text{sd}(\hat{z}_k)} \left\{ \beta_{jk0} - \sum_{h=1}^q \frac{\bar{\mathbf{u}}_h}{\text{sd}(\mathbf{u}_h)} \beta_{jkh} \right\}$ and $\tilde{\beta}_{jkh} = \frac{\beta_{jkh}}{\text{sd}(\mathbf{u}_h) \text{sd}(\hat{z}_k)}$. Notice $\tilde{\beta}_{jkh}$ and β_{jkh} only differs in scale by a positive scalar. Therefore, parameter estimates can be interpreted with non-standardized covariates after a re-calculation as in (S1).

S1.6 Positive definiteness

As our goal is to identify edges that are modulated by external covariates, we choose to employ the computationally efficient node-wise regressions and bypass the need to work with the entire covariance/precision matrix. If the true covariance/precision matrix is positive

	$p = 25$ $q = 50$	$p = 25$ $q = 100$	$p = 50$ $q = 50$	$p = 50$ $q = 100$
β_{error}	1.378 (0.006)	1.417 (0.006)	2.228 (0.005)	2.291 (0.005)
β_{error}^r	1.372 (0.005)	1.446 (0.006)	2.226 (0.005)	2.292 (0.004)

Table S5: Estimation accuracy of β_j 's and β_j^r 's in simulations with $n = 200$, varying network size p and covariate dimension q .

definite, it then follows from Theorems 1-4 that the estimated precision is asymptotically positive definite. However, for finite sample cases, it may be desirable to ensure the positive definiteness of the final estimator. We find that, assuming the true precision matrix is diagonal dominant and the covariates are in known and bounded intervals, we can adopt a post-hoc re-scaling step that gives the same estimator asymptotically as guaranteed by Theorems 1-4. Specifically, without loss of generality, assume $u_h \in [0, 1]$ (if not, rescale u_h first). For any j such that $\|\hat{\beta}_j\|_1 > 1$, we set the final estimate to $[\hat{\mathbf{B}}_h]_{j\cdot}/\hat{\sigma}^{jj}$ and $[\hat{\mathbf{B}}_h]_{\cdot j}/\hat{\sigma}^{jj}$.

We have evaluated the performance of the above procedure under the same setting as in Section 6. Recall in our simulations, the true precision matrix is diagonal dominant and $u_h \in [0, 1]$. The results over 200 data replicates are reported in Table S5, where the rescaled $\hat{\beta}_j$ is denoted as $\hat{\beta}_j^r$. Table S5 suggests that the original and rescaled estimators have similar estimation errors.

S2 Technical Lemmas

We state the technical lemmas that will be used in our proofs.

Lemma 1 (Lemma 1 in Bellec et al. (2018)). *Let $\text{pen} : \mathbb{R}^d \rightarrow \mathbb{R}$ be any convex function and $\hat{\beta}$ be defined by*

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^d} \left\{ \|\mathbf{y} - \mathbf{W}\beta\|_2^2 + \text{pen}(\beta) \right\},$$

where $\mathbf{W} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$. Then for $\boldsymbol{\beta} \in \mathbb{R}^d$,

$$\|\mathbf{y} - \mathbf{W}\hat{\boldsymbol{\beta}}\|_2^2 + \text{pen}(\hat{\boldsymbol{\beta}}) + \|\mathbf{W}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|_2^2 \leq \|\mathbf{y} - \mathbf{W}\boldsymbol{\beta}\|_2^2 + \text{pen}(\boldsymbol{\beta}).$$

Lemma 2 (Theorem F in [Graybill and Marsaglia \(1957\)](#)). Let $\boldsymbol{\epsilon}_j \sim \mathcal{N}_p(\mathbf{0}, \sigma^2 \mathbf{I})$ and $\mathbf{A}$ be an $p \times p$ idempotent matrix with rank equals to $r \leq p$. Then, $\boldsymbol{\epsilon}_j^\top \mathbf{A} \boldsymbol{\epsilon}_j / \sigma^2$ follows a χ^2 distribution with r degrees of freedom.

Lemma 3 (Lemma 1 in [Laurent and Massart \(2000\)](#)). Suppose that U follows a χ^2 distribution with r degrees of freedom. For any $x > 0$, it holds that

$$P(U - r \geq 2\sqrt{rx} + 2x) \leq \exp(-x).$$

Lemma 4 (Proposition 5.16 in [Vershynin \(2010\)](#)). Let $X_1, \dots, X_n$ be independent mean zero sub-exponential random variables. Let $v_1 = \max_i \|X_i\|_{\psi_1}$, where $\|X_i\|_{\psi_1} = \sup_{d \geq 1} d^{-1} (E|X_i|^d)^{1/d}$ denotes the sub-exponential norm. There exists a constant c such that, for any $t > 0$,

$$P\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left\{-c \min\left(\frac{t^2}{v_1^2 n}, \frac{t}{z_1}\right)\right\}.$$

Lemma 5. Consider independent vectors $(y_1, \mathbf{x}_1), \dots, (y_n, \mathbf{x}_n)$ in $\mathbb{R} \times \mathbb{R}^p$ such that $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i$ and $\mathbf{x}_i$ is elementwise sub-Gaussian, $i \in [n]$. Let ϵ_i 's be independent Gaussian errors with non-constant variances and assume that $\sup_{i \in [n]} \text{Var}(\epsilon_i)$ is bounded by a constant $K_1 > 0$. Suppose $\|\boldsymbol{\beta}\|_0 = k$ and $\lambda_{\min}(\text{Cov}(\mathbf{x}_1)) \geq 1/\phi_0$ for some $\phi_0 > 0$. Let

$$\hat{\boldsymbol{\beta}}_\lambda = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^p} \frac{1}{2n} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2 + \lambda \|\boldsymbol{\theta}\|_1.$$

When $\lambda = 14\sigma_n \sqrt{\tau_1 \log p / n} + \tau_1 K (\log p \log n)^{1/2} / n$ for any $\tau_1 > 0$, the lasso estimate satisfies with probability at least $1 - 3p^{-\tau_1}$ that

$$\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|_2 \lesssim \sigma_n \sqrt{\frac{k \log p}{n}} + \frac{k^{1/2} \log p}{n},$$

where $\sigma_n = \frac{1}{n} \sum_{i=1}^n \text{Var}(\epsilon_i)$.

The above result is adapted from Theorem 4.5 in [Kuchibhotla and Chakraborty \(2018\)](#) by considering Gaussian errors.

Lemma 6 (Theorem 4.1 in [Kuchibhotla and Chakraborty \(2018\)](#)). *Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be independent random vectors in $\mathbb{R}^p$. Assume each element of $\mathbf{X}_i$ is sub-exponential with $\|X_{i,j}\|_{\psi_1} < K_2$, $i \in [n], j \in [p]$. where $\|X_{i,j}\|_{\psi_1} = \sup_{d \geq 1} d^{-1}(E|X_{i,j}|^d)^{1/d}$ denotes the sub-exponential norm. Let $\hat{\Sigma}_{\mathbf{X}} = \mathbf{X}^\top \mathbf{X}/n$ and $\Sigma_{\mathbf{X}} = \mathbb{E}(\mathbf{X}^\top \mathbf{X}/n)$. Define*

$$\Upsilon_{n,k} = \max_{j,k} \frac{1}{n} \sum_{i=1}^n \text{Var}\{X_{i,j}X_{i,k}\}.$$

Then for any $t > 0$, with probability at least $1 - \mathcal{O}(p^{-1})$,

$$\sup_{\|\mathbf{v}\|_0 \leq k, \|\mathbf{v}\|_2 \leq 1} \left| \mathbf{v}^\top (\hat{\Sigma}_{\mathbf{X}} - \Sigma_{\mathbf{X}}) \mathbf{v} \right| \lesssim k \sqrt{\frac{\Upsilon_{n,k} \log p}{n}} + K_2^2 \frac{k(\log n \log p)^2}{n}.$$

Lemma 7 (Lemma 12 in [Loh and Wainwright \(2011\)](#)). *For any symmetric matrix $\Sigma \in \mathbb{R}^{p \times p}$ and if $|\mathbf{v}^\top \Sigma \mathbf{v}| \leq \delta_1$ for any $\mathbf{v} \in \{\mathbf{v} : \|\mathbf{v}\|_0 \leq 2s \text{ and } \|\mathbf{v}\|_2 = 1\}$, then*

$$|\mathbf{v}^\top \Sigma \mathbf{v}| \leq 27\delta_1(\|\mathbf{v}\|_2^2 + \frac{1}{s}\|\mathbf{v}\|_1^2), \text{ for any } \mathbf{v} \in \mathbb{R}^p.$$

S3 Proofs of Main Results

We begin by recalling the notation used throughout the paper. We denote the true parameters by β_j , $j \in [p]$, though, in some places and without ambiguities, we use them to denote the corresponding parameters or the arguments in functions. The index set $\{1, \dots, (p-1)(q+1)\}$ is partitioned into $q+1$ groups, indexed by $(0), (1), \dots, (q) \subset \{1, \dots, (p-1)(q+1)\}$. For a group index subset $\mathcal{G} \in \{1, \dots, q\}$, we let $(\mathcal{G}) = \cup_{h \in \mathcal{G}}(h)$, $(\mathcal{G}^c) = \cup_{h \notin \mathcal{G}}(h)$, and $(\beta_j)_{(\mathcal{G})}$ represent a sub-vector of β_j indexed by $(\mathcal{G})$. We define $\|\beta_{j,-0}\|_{1,2} = \sum_{h=1}^q \|(\beta_j)_{(h)}\|_2$. We use $\mathcal{S}_j$ to denote the element-wise support set of β_j , i.e., $\mathcal{S}_j = \{l : (\beta_j)_l \neq 0, l \in [(p-1)(q+1)]\}$, and $\mathcal{G}_j$ to denote the group-wise support set of β_j , i.e., $\mathcal{G}_j = \{h : (\beta_j)_{(h)} \neq \mathbf{0}, h \in [q]\}$. Moreover, we let $s_j = |\mathcal{S}_j|$ and $s_{j,g} = |\mathcal{G}_j|$.

S3.1 Proof of Theorem 1

As $\hat{\beta}_j$ is a minimizer of the objective function (9) and the sparse group penalty function in (9) is convex, Lemma 1 implies that

$$\begin{aligned} & \frac{1}{2n} \|\mathbf{z}_j - \mathbf{W}\hat{\beta}_j\|_2^2 + \lambda \|\hat{\beta}_j\|_1 + \lambda_g \|\hat{\beta}_{j,-0}\|_{1,2} + \frac{1}{2n} \|\mathbf{W}(\hat{\beta}_j - \beta_j)\|_2^2 \\ & \leq \frac{1}{2n} \|\mathbf{z}_j - \mathbf{W}\beta_j\|_2^2 + \lambda \|\beta_j\|_1 + \lambda_g \|\beta_{j,-0}\|_{1,2}. \end{aligned}$$

Writing $\Delta = \hat{\beta}_j - \beta_j$ and reorganizing terms in the above inequality gives

$$\frac{1}{n} \|\mathbf{W}\Delta\|_2^2 + \lambda \|\hat{\beta}_j\|_1 + \lambda_g \|\hat{\beta}_{j,-0}\|_{1,2} \leq \frac{1}{n} \langle \epsilon_j, \mathbf{W}\Delta \rangle + \lambda \|\beta_j\|_1 + \lambda_g \|\beta_{j,-0}\|_{1,2}.$$

Using the fact that $\|\hat{\beta}_j\|_1 = \|(\hat{\beta}_j)_{\mathcal{S}_j}\|_1 + \|(\hat{\beta}_j)_{\mathcal{S}_j^c}\|_1$, $\|\beta_j\|_1 = \|(\beta_j)_{\mathcal{S}_j}\|_1$, $\|\hat{\beta}_j\|_{1,2} = \|(\hat{\beta}_j)_{(\mathcal{G}_j)}\|_{1,2} + \|(\hat{\beta}_j)_{(\mathcal{G}_j^c)}\|_{1,2}$, $\|\beta_j\|_{1,2} = \|(\beta_j)_{(\mathcal{G}_j)}\|_{1,2}$ and applying the triangle inequalities of $\|\cdot\|_1$ and $\|\cdot\|_{1,2}$, we arrive at

$$\frac{1}{n} \|\mathbf{W}\Delta\|_2^2 + \lambda \|\Delta_{\mathcal{S}_j^c}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j^c)}\|_{1,2} \leq \frac{1}{n} \langle \epsilon_j, \mathbf{W}\Delta \rangle + \lambda \|\Delta_{\mathcal{S}_j}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j)}\|_{1,2}. \quad (\text{S2})$$

Defining $\hat{\mathcal{S}}_j = \{l : (\hat{\beta}_j)_l \neq 0, l \in [(p-1)(q+1)]\}$ and letting $\tilde{\mathcal{S}}_j = \mathcal{S}_j \cup \hat{\mathcal{S}}_j$, we obtain

$$\begin{aligned} \langle \epsilon_j, \mathbf{W}\Delta \rangle &= \langle \epsilon_j, \mathcal{P}_{\tilde{\mathcal{S}}_j} \mathbf{W}_{\tilde{\mathcal{S}}_j} \Delta_{\tilde{\mathcal{S}}_j} \rangle \\ &= \langle \mathcal{P}_{\tilde{\mathcal{S}}_j} \epsilon_j, \mathbf{W}\Delta \rangle \leq \frac{1}{2a_1} \|\mathbf{W}\Delta\|_2^2 + \frac{a_1}{2} \|\mathcal{P}_{\tilde{\mathcal{S}}_j} \epsilon_j\|_2^2, \end{aligned} \quad (\text{S3})$$

where the second equality holds with $\tilde{\mathcal{S}}_j = \mathcal{S}_j \cup \hat{\mathcal{S}}_j$ and the last inequality comes from that $2ab \leq ta^2 + b^2/t$ for any $t > 0$. Here, $\mathcal{P}_{\tilde{\mathcal{S}}_j}$ is the orthogonal projection matrix onto the column space of $\mathbf{W}_{\tilde{\mathcal{S}}_j}$. As opposed to the classic techniques that bound $\langle \epsilon, \mathbf{W}\Delta \rangle$ with $\|\mathbf{W}^\top \epsilon\|_\infty \|\Delta\|_1$ or $\|\mathbf{W}^\top \epsilon\|_{\infty,2} \|\Delta\|_{1,2}$ (Bickel et al., 2009; Lounici et al., 2011; Negahban et al., 2012), we bound this term in (S3) with $\|\mathbf{W}\Delta\|_2^2/(2a_1) + a_1 \|\mathcal{P}_{\tilde{\mathcal{S}}_j} \epsilon_j\|_2^2/2$, which is useful in our proof to more sharply bound the term $\|\mathcal{P}_{\tilde{\mathcal{S}}_j} \epsilon_j\|_2^2$. This indeed is a challenging step as the group lasso penalty term in (9) is not decomposable with respect to $\mathcal{S}_j$ and, hence,

the existing techniques based on decomposable regularizers and null space properties are non-applicable. We provide a new proof, which is divided into three steps.

Step 1: We first bound $\|\mathcal{P}_{\mathcal{S}_j}\epsilon_j\|_2^2$ with some cardinality measures. Given any $\mathcal{J} \subset [(p-1)(q+1)]$ and $\gamma \in \{0,1\}^{(p-1)(q+1)}$ satisfying $\gamma_{\mathcal{J}} = \mathbf{1}$ and $\gamma_{\mathcal{J}^c} = \mathbf{0}$, we write $\mathcal{G}(\mathcal{J}) = \{h : (\gamma)_{(h)} \neq \mathbf{0}, h \in [q]\}$. In this step, we aim to show that, given $0 \leq s_{j,g} \leq q+1$ and $0 \leq s_j \leq (p-1)(q+1)$, the following holds

$$\mathbb{P} \left[\sup_{|\mathcal{J}|=s_j, |\mathcal{G}(\mathcal{J})|=s_{j,g}} \|\mathcal{P}_{\mathcal{J}}\epsilon_j\|_2^2 \geq 6\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\} + t\sigma_{\epsilon_j}^2 \right] \leq c_1 \exp(-c_2 t),$$

where c_1, c_2 are positive constants.

As the projection matrix $\mathcal{P}_{\mathcal{J}}$ is idempotent, Lemma 2 implies that

$$\|\mathcal{P}_{\mathcal{J}}\epsilon_j\|_2^2 / \sigma_{\epsilon_j}^2 \sim \chi_d^2, \quad d < s_j,$$

where d is the rank of $\mathcal{P}_{\mathcal{J}}$. Next, we find the size of $\{\mathcal{J} \subset [(p-1)(q+1)], |\mathcal{J}| = s_j, |\mathcal{G}(\mathcal{J})| = s_{j,g}\}$ by considering two cases (i) $s_{j,g} = s_j$ and (ii) $s_{j,g} < s_j$. These are the only two possible cases since $\mathcal{S}_j$ includes all nonzero elements in $(\beta_j)_{(0)}$ and $(\beta_j)_{(1)}, \dots, (\beta_j)_{(q)}$.

case (i): $s_{j,g} = s_j$. Here, $\{\mathcal{J} \subset [(p-1)(q+1)], |\mathcal{J}| = s_j, |\mathcal{G}(\mathcal{J})| = s_{j,g}\}$ contains $\binom{q}{s_{j,g}}(p-1)^{s_j}$ elements. It follows from Stirling's approximation that $\log \binom{q}{s_{j,g}} \leq s_{j,g} \log(eq/s_{j,g})$. Therefore, $\log \left\{ \binom{q}{s_{j,g}}(p-1)^{s_j} \right\} \leq s_j \log p + s_{j,g} \log(eq/s_{j,g})$.

case (ii): $s_{j,g} < s_j$. The number of elements in $\{\mathcal{J} \subset [(p-1)(q+1)], |\mathcal{J}| = s_j, |\mathcal{G}(\mathcal{J})| = s_{j,g}\}$ is bounded above by $\binom{q}{s_{j,g}} \binom{(p-1)+(p-1)s_{j,g}}{s_j}$. By Stirling's approximation, we have $\log \binom{(p-1)+(p-1)s_{j,g}}{s_j} \leq s_j \log(e(p-1)(s_{j,g}+1)/s_j) \leq s_j \log(ep)$. Therefore, we have $\log \left\{ \binom{q}{s_{j,g}} \binom{(p-1)+(p-1)s_{j,g}}{s_j} \right\} \leq s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})$.

Combining these two cases, we conclude that $|\{\mathcal{J} \subset [(p-1)(q+1)], |\mathcal{J}| = s_j, |\mathcal{G}(\mathcal{J})| = s_{j,g}\}|$ is bounded above by $s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})$. Applying Lemma 3 and applying the union bound leads to the desired result in Step 1.

Step 2: Using the result from Step 1, we find an upper bound for $\|\mathcal{P}_{\hat{\mathcal{S}}_j} \epsilon_j\|_2^2$. First, we define

$$r_{s,s_g} = \left(\sup_{|\mathcal{J}|=s, |\mathcal{G}(\mathcal{J})|=s_g} \|\mathcal{P}_{\mathcal{J}} \epsilon_j\|_2^2 - M\sigma_{\epsilon_j}^2 \{s \log(ep) + s_g \log(eq/s_g)\} \right)_+,$$

and $r = \sup_{1 \leq s \leq (p-1)(q+1), 0 \leq s_g \leq q} r_{s,s_g}$, where $M > 0$ is a constant to be specified later. Then,

$$\begin{aligned} \|\mathcal{P}_{\hat{\mathcal{S}}_j} \epsilon_j\|_2^2 &\leq M\sigma_{\epsilon_j}^2 \{\tilde{s}_j \log(ep) + \tilde{s}_{j,g} \log(eq/s_{j,g})\} + r \\ &\leq M\sigma_{\epsilon_j}^2 \{(s_j + \hat{s}_j) \log(ep) + (s_{j,g} + \hat{s}_{j,g}) \log(eq/s_{j,g})\} + r. \end{aligned} \quad (\text{S4})$$

With $M = 9$, the result from Step 1 gives

$$\begin{aligned} \mathbb{P}\{r \geq t\sigma_{\epsilon_j}^2\} &\leq \sum_{s=1}^{(p-1)(q+1)} \sum_{s_g=0}^q \mathbb{P}\{r_{s,s_g} \geq t\sigma_{\epsilon_j}^2\} \\ &\leq \sum_{s=1}^{(p-1)(q+1)} \sum_{s_g=0}^q c_1 \exp[-c_2 t - 3c_2 \{s \log(ep) + s_g \log(eq/s_g)\}]. \end{aligned}$$

Step 3: This step derives an inequality for $\|\mathcal{P}_{\hat{\mathcal{S}}_j} \epsilon_j\|_2^2$ by utilizing the computational optimality of $\hat{\beta}_j$. Since the objective function is convex, $\hat{\beta}_j$ is a stationary point of

$$\frac{1}{2n} \|\mathbf{z}_j - \mathbf{W}\beta_j\|_2^2 + \lambda \|\beta_j\|_1 + \lambda_g \|\beta_{j,-0}\|_{1,2}.$$

By the KKT conditions, for any $l \in \hat{\mathcal{S}}_j \cap (0)$, $h \in [q]$, $\hat{\beta}_j$ must satisfy that

$$\lambda \text{sign}\{(\hat{\beta}_j)_l\} = \frac{1}{n} \langle \mathbf{w}_l, \mathbf{z}_j - \mathbf{W}\hat{\beta}_j \rangle. \quad (\text{S5})$$

Similarly, for any $l \in \hat{\mathcal{S}}_j \cap (h)$, it must satisfy that

$$\lambda \text{sign}\{(\hat{\beta}_j)_l\} + \lambda_g \frac{(\hat{\beta}_j)_l}{\|(\hat{\beta}_j)_{(h)}\|_2^2} = \frac{1}{n} \langle \mathbf{w}_l, \mathbf{z}_j - \mathbf{W}\hat{\beta}_j \rangle. \quad (\text{S6})$$

Squaring both sides of (S5) and (S6) and summing over all $l \in \hat{\mathcal{S}}_j$ gives

$$\lambda^2 \hat{s}_j + \lambda_g^2 \hat{s}_{j,g} \leq \frac{1}{n^2} \|\mathbf{W}_{\hat{\mathcal{S}}}^\top (\mathbf{z}_j - \mathbf{W}\hat{\beta}_j)\|_2^2,$$

due to $\text{sign}\{(\hat{\beta}_j)_l\}(\hat{\beta}_j)_l \geq 0$.

Consider $\mathbf{W}_i^\top \mathbf{v}$, where $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_p)$, $\mathbf{v}_l \in \mathbb{R}^{q+1}$ and $\|\mathbf{v}\| = 1$. We have $\mathbf{W}_i = (z_1^{(i)}, z_1^{(i)} u_1^{(i)}, \dots, z_1^{(i)} u_q^{(i)}, \dots, z_p^{(i)}, z_p^{(i)} u_1^{(i)}, \dots, z_p^{(i)} u_q^{(i)})$. With a slight abuse of notation, we include the intercept term into $\mathbf{u}^{(i)}$ in the subsequent development. Letting $\mathbf{V} = [\mathbf{v}_1^\top, \dots, \mathbf{v}_p^\top] \in \mathbb{R}^{(q+1) \times p}$, we can reexpress $\mathbf{W}_i \mathbf{v}$ as $\mathbf{W}_i \mathbf{v} = \mathbf{u}^{(i)\top} \mathbf{V} \mathbf{z}^{(i)}$, $i \in [n]$. Consequently, by the law of total expectation and Assumption 1, we have

$$\begin{aligned}
\mathbb{E} \left(\mathbf{v}^\top \frac{\mathbf{W}^\top \mathbf{W}}{n} \mathbf{v} \right) &= \frac{1}{n} \sum_{i=1}^n \mathbb{E} \left(\mathbf{u}^{(i)\top} \mathbf{V} \mathbf{z}^{(i)} \right)^2 \\
&= \mathbb{E} \left[\mathbb{E} \left\{ \frac{1}{n} \sum_{i=1}^n \mathbb{E} \left(\mathbf{u}^{(i)\top} \mathbf{V} \mathbf{z}^{(i)} \right)^2 \middle| \{\mathbf{u}^{(i)}\}_{i \in [n]} \right\} \right] \\
&= \mathbb{E} \left\{ \frac{1}{n} \sum_{i=1}^n \mathbf{u}^{(i)\top} \mathbf{V} \Sigma(\mathbf{u}^{(i)}) \mathbf{V}^\top \mathbf{u}^{(i)} \right\} \\
&\geq \phi_1 \times \mathbb{E} \left\{ \text{tr} \left(\mathbf{u}^{(1)\top} \mathbf{V} \mathbf{V}^\top \mathbf{u}^{(1)} \right) \right\} \\
&\geq \phi_1 \times \lambda_{\min}(\text{Cov}(\mathbf{u}^{(1)})) \text{tr}(\mathbf{V} \mathbf{V}^\top) = \phi_1 / \phi_0,
\end{aligned} \tag{S7}$$

where we have used the fact $\text{tr}(\mathbf{A}\mathbf{B}) \geq \lambda_{\min}(\mathbf{A})\text{tr}(\mathbf{B})$ for positive semi-definite matrices $\mathbf{A}$ and $\mathbf{B}$ and $\text{tr}(\mathbf{V} \mathbf{V}^\top) = 1$. By the Cauchy–Schwarz inequality, we deduce

$$\max_{j,k} \frac{1}{n} \sum_{i=1}^n \mathbb{E} \{ (\mathbf{W}_{ij} \mathbf{W}_{ik})^2 \} = \max_{l_1, l_2, l_3, l_4} \mathbb{E} \left(z_{l_1}^{(1)2} z_{l_2}^{(1)2} u_{l_3}^{(1)2} u_{l_4}^{(1)2} \right) = \mathcal{O}(1), \tag{S8}$$

where we have used the fact that $z_{l_1}^{(i)}$ and $u_{l_2}^{(i)}$ have bounded eighth moments, as they are both sub-Gaussian with a bounded sub-Gaussian norm. It then follows from Lemma 6 and with (S7) and Assumption 3 that, with probability at least $1 - C_0 \exp\{-(\log p + \log q)\}$, we have $\|\mathbf{W}_{\hat{\mathcal{S}}}\|^2/n \leq M_1$ for some $M_1 > 0$. Since $\hat{\mathcal{S}}_j \in \tilde{\mathcal{S}}$, it holds that

$$\lambda^2 \hat{s}_j + \lambda_g^2 \hat{s}_{j,g} \leq \frac{2M_1}{n} \|\mathbf{W} \Delta\|_2^2 + \frac{2M_1}{n} \|\mathcal{P}_{\tilde{\mathcal{S}}_j} \boldsymbol{\epsilon}_j\|_2^2. \tag{S9}$$

Combining (S4) and (S9) and letting

$$\lambda = C \sigma_{\epsilon_j} \sqrt{\log(ep)/n + s_{j,g} \log(eq/s_{j,g})/(ns_j)}, \quad \lambda_g = \sqrt{s_j/s_{j,g}} \lambda,$$

where $C = 3(M_1 a_2)^{1/2}$ for some $a_2 > 0$, we arrive at

$$(1 - \frac{2}{a_2}) \|\mathcal{P}_{\tilde{s}_j} \epsilon_j\|_2^2 \leq 9\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\} + \frac{2}{a_2} \|\mathbf{W} \Delta\|_2^2 + r. \quad (\text{S10})$$

This, together with (S2) and (S3), implies

$$\begin{aligned} & \frac{\|\mathbf{W} \Delta\|_2^2}{n} + \lambda \|\Delta_{\mathcal{S}_j^c}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j^c)}\|_{1,2} \\ & \leq \frac{1}{2a_1} \frac{\|\mathbf{W} \Delta\|_2^2}{n} + \frac{9a_1 a_2}{2(a_2 - 2)} \frac{\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}}{n} \\ & \quad + \frac{a_1}{a_2 - 2} \frac{\|\mathbf{W} \Delta\|_2^2}{n} + \frac{a_1 a_2}{2(a_2 - 2)n} r + \lambda \|\Delta_{\mathcal{S}_j}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j)}\|_{1,2}. \end{aligned} \quad (\text{S11})$$

Next, we have that

$$\frac{\|\Delta_{\mathcal{S}_j}\|_1}{\sqrt{s_j}} + \frac{\|\Delta_{(\mathcal{G}_j)}\|_{1,2}}{\sqrt{s_{j,g}}} \leq \|\Delta_{\mathcal{S}_j}\|_2 + \|\Delta_{(\mathcal{G}_j)}\|_2 \leq 2 \frac{\phi_1}{\phi_0} \|\Sigma_{\mathbf{W}}^{1/2} \Delta\|_2,$$

where the first inequality is due to that $\|\Delta_{\mathcal{S}_j}\|_1 \leq \sqrt{s_j} \|\Delta_{\mathcal{S}_j}\|_2$, $\|\Delta_{(\mathcal{G}_j)}\|_{1,2} \leq \sqrt{s_{j,g}} \|\Delta_{(\mathcal{G}_j)}\|_2$ and the second inequality holds because $\|\Delta_{\mathcal{S}_j}\|_2 + \|\Delta_{(\mathcal{G}_j)}\|_2 < 2\|\Delta\|_2$ trivially with $\Delta_{\mathcal{S}_j}$ and $\Delta_{(\mathcal{G}_j)}$ being the sub-vectors of Δ , and $\lambda_{\min}(\Sigma_{\mathbf{W}}) \geq \phi_1/\phi_0 > 0$ in (S7). Consequently,

$$\lambda \|\Delta_{\mathcal{S}_j}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j)}\|_{1,2} \leq 2C \frac{\phi_1}{\phi_0} \sqrt{e_j} \|\Sigma_{\mathbf{W}}^{1/2} \Delta\|_2 \leq a_3 C \frac{\phi_1}{\phi_0} e_j + \frac{1}{a_3} \|\Sigma_{\mathbf{W}}^{1/2} \Delta\|_2^2,$$

where $e_j = \sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}/n$ and the last inequality comes from that $2ab \leq ta^2 + b^2/t$ for any $t > 0$. Plugging this into (S11), we obtain

$$\begin{aligned} & \left\{ 1 - \frac{1}{2a_1} - \frac{a_1}{a_2 - 2} \right\} \frac{\|\mathbf{W} \Delta\|_2^2}{n} \\ & \leq \left\{ \frac{9a_1 a_2}{2(a_2 - 2)} + C a_3 \frac{\phi_1}{\phi_0} \right\} \frac{\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}}{n} + \frac{1}{a_3} \|\Sigma_{\mathbf{W}}^{1/2} \Delta\|_2^2 + \frac{a_1 a_2}{2(a_2 - 2)n} r. \end{aligned} \quad (\text{S12})$$

It remains to bound the distance between $\|\mathbf{W} \Delta\|_2^2/n$ and $\|\Sigma_{\mathbf{W}}^{1/2} \Delta\|_2^2$. To proceed, we first show with probability at least $1 - C' \exp\{-(\log p + \log q)\}$,

$$\sup_{\mathbf{v} \in \mathbb{K}_0(2C_{\beta_j, s_j})} \left| \mathbf{v}^\top \left(\frac{\mathbf{W}^\top \mathbf{W}}{n} - \Sigma_{\mathbf{W}} \right) \mathbf{v} \right| \leq 1/L, \quad (\text{S13})$$

where L is an arbitrarily large constant and $\mathbb{K}_0(2C_{\beta_j}s_j) = \{\mathbf{v} : \|\mathbf{v}\|_0 \leq 2C_{\beta_j}s_j \text{ and } \|\mathbf{v}\|_2 = 1\}$ for some positive constant C_{β_j} .

Given (S8), it follows from Lemma 6 and Assumption 3 that with probability at least $1 - C' \exp\{-(\log p + \log q)\}$

$$\left| \mathbf{v}^\top \left(\frac{\mathbf{W}^\top \mathbf{W}}{n} - \boldsymbol{\Sigma}_{\mathbf{W}} \right) \mathbf{v} \right| = o(1).$$

Combing this with the result in Lemma 7, we have, with probability at least $1 - C' \exp\{-(\log p + \log q)\}$,

$$\left| \Delta^\top \left(\frac{\mathbf{W}^\top \mathbf{W}}{n} - \boldsymbol{\Sigma}_{\mathbf{W}} \right) \Delta \right| \leq \frac{1}{L'} \left(\|\Delta\|_2^2 + \frac{1}{s_j} \|\Delta\|_1^2 \right), \quad (\text{S14})$$

where L' is an arbitrarily large positive constant. Plugging (S14) into (S12) and choosing proper constants a_1 , a_2 and a_3 (e.g., $a_1 = 2$, $a_2 = 6$, $a_3 = 6$), we have

$$\frac{1}{2} \|\boldsymbol{\Sigma}_{\mathbf{W}}^{1/2} \Delta\|_2^2 \lesssim \frac{\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}}{n} + \frac{1}{L'} \left(\|\Delta\|_2^2 + \frac{1}{s_j} \|\Delta\|_1^2 \right) + \frac{\sigma_{\epsilon_j}^2}{n}, \quad (\text{S15})$$

with probability at least $1 - c_1 \exp[-c'_2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}]$, due to that

$$\mathbb{P} \left[r \geq M_0 \sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\} \right] \leq c_1 \exp[-c'_2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}],$$

for a large positive constant M_0 . Next, taking $a_1 = 2 - \sqrt{2}$ and $a_2 = 6$ in (S11) and using the expressions for λ , λ_g , we have with probability at least $1 - c_1 \exp[-c'_2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}]$ that

$$\frac{\|\Delta_{\mathcal{S}_j^c}\|_1}{\sqrt{s_j}} + \frac{\|\Delta_{(\mathcal{G}_j^c)}\|_{1,2}}{\sqrt{s_{j,g}}} \leq \sigma_{\epsilon_j} \sqrt{\frac{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})}{n}} + \frac{\|\Delta_{\mathcal{S}_j}\|_1}{\sqrt{s_j}} + \frac{\|\Delta_{(\mathcal{G}_j)}\|_{1,2}}{\sqrt{s_{j,g}}}. \quad (\text{S16})$$

Adding $\|\Delta_{\mathcal{S}_j}\|_1/\sqrt{s_j}$ to both sides of (S16), we get

$$\frac{\|\Delta\|_1}{\sqrt{s_j}} \leq \sqrt{e_j} + 3\|\Delta\|_2. \quad (\text{S17})$$

Plugging (S17) into (S15) and with $\lambda_{\min}(\boldsymbol{\Sigma}_{\mathbf{W}}) \geq \phi_1/\phi_0 > 0$ in (S7), we have

$$\|\Delta\|_2^2 \lesssim \frac{\sigma_{\epsilon_j}^2}{n} \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\} + \frac{\sigma_{\epsilon_j}^2}{n},$$

with probability at least $1 - C_1 \exp[-C_2\{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}]$, for some positive constants C_1, C_2 .

□

S3.2 Proof of Theorem 2

We first establish the element-wise error bound of $\hat{\beta}_j$ in (14), which is to be accomplished in three steps.

Step 1: With $\Psi = \mathbf{W}^\top \mathbf{W}/n$, this step shows that with probability at least $1 - 2 \exp\{-c_3(\log p + \log q)\}$,

$$\|\Psi \Delta\|_\infty \leq \frac{3\eta_j \lambda}{2} \quad (\text{S18})$$

for a constant $c_3 > 0$. We also show with probability at least $1 - 2 \exp\{-c'_3 \log p\}$ that

$$\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j \|\Delta_{\mathcal{S}_j}\|_1. \quad (\text{S19})$$

The KKT conditions state that $\boldsymbol{\theta}$, an optimizer of (9), satisfies

$$\begin{cases} (\mathbf{W}^\top (\mathbf{z}_j - \mathbf{W}\boldsymbol{\theta})/n)_l = \text{sign}(\theta_l) \lambda & \text{if } \theta_l \neq 0, l \in (0) \\ (\mathbf{W}^\top (\mathbf{z}_j - \mathbf{W}\boldsymbol{\theta})/n)_l = \text{sign}(\theta_l) \lambda + \lambda_g \frac{\theta_l}{\|\boldsymbol{\theta}_{(h)}\|_2} & \text{if } \theta_l \neq 0, l \in (h) \\ |(\mathbf{W}^\top (\mathbf{z}_j - \mathbf{W}\boldsymbol{\theta})/n)_l| < \eta_j \lambda & \text{if } \theta_l = 0. \end{cases}$$

Thus, $\hat{\beta}_j$ must satisfy that

$$\left\| \frac{1}{n} \mathbf{W}^\top (\mathbf{z}_j - \mathbf{W}\hat{\beta}_j) \right\|_\infty \leq \eta_j \lambda.$$

If we can show that with high probability

$$\frac{1}{n} \|\mathbf{W}^\top \boldsymbol{\epsilon}_j\|_\infty \leq \frac{\eta_j \lambda}{2}, \quad (\text{S20})$$

we conclude that (S18) holds with at least the same probability.

To prove (S20), we first define $V_l = \mathbf{w}_l^\top \boldsymbol{\epsilon}_j/n$, $l \in [(p-1)(q+1)]$, where $\mathbf{w}_l$ is the l th column of $\mathbf{W}$. As $\mathbf{u}_h^{(i)}$ is bounded as assumed in Assumption 1, $w_l^{(i)}$ is sub-Gaussian. Hence,

V_l is a sum of independent sub-exponential random variables and $\text{Var}(w_l^{(i)}) \leq M^2 \phi_2$ for some $M > 0$. Applying Lemma 4 gives that

$$\mathbb{P}\left(|V_l| > \frac{\eta_j \lambda}{2}\right) \leq 2 \exp\left(-\frac{c\eta_j^2 \lambda^2 n}{4M^2 \phi_2 \sigma_{\epsilon_j}^2}\right) \leq 2 \exp\{-C'_0(\log p + \log q)\}, \quad (\text{S21})$$

where $C'_0 = cC^2/(4M^2 \phi_2)$, and C is as defined in (12), and the last inequality is due to

$$\eta_j \lambda \geq C\sigma_{\epsilon_j} \sqrt{\frac{\log p}{n}} + C\sigma_{\epsilon_j} \sqrt{\frac{\log q}{n}}.$$

With constant C in (12) chosen sufficiently large, we have $C'_0 > 1$. Applying the union bound inequality gives

$$\begin{aligned} \mathbb{P}\left(\frac{1}{n} \|\mathbf{W}^\top \boldsymbol{\epsilon}_j\|_\infty \geq \frac{\eta_j \lambda}{2}\right) &\leq \mathbb{P}\left(\max_l |V_l| \geq \frac{\eta_j \lambda}{2}\right) \\ &\leq 2 \exp\{-(C'_0 - 1)(\log p + \log q)\}. \end{aligned}$$

We have finished showing (S18) of Step 1 by taking $c_3 = C'_0 - 1$.

Next, we show that $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 3\eta_j \|\Delta_{\mathcal{S}_j}\|_1$ with probability greater than $1 - \exp\{-c'_3(\log p)\}$.

The definition of $\hat{\boldsymbol{\beta}}_j$ implies that

$$\frac{1}{2n} \|\mathbf{z}_j - \mathbf{W} \hat{\boldsymbol{\beta}}_j\|_2^2 + \lambda \|\hat{\boldsymbol{\beta}}_j\|_1 + \lambda_g \|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2} \leq \frac{1}{2n} \|\boldsymbol{\epsilon}_j\|_2^2 + \lambda \|\boldsymbol{\beta}_j\|_1 + \lambda_g \|\boldsymbol{\beta}_{j,-0}\|_{1,2}.$$

Developing the left hand side of the above inequality gives

$$\lambda \|\hat{\boldsymbol{\beta}}_j\|_1 + \lambda_g \|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2} \leq \lambda \|\boldsymbol{\beta}_j\|_1 + \lambda_g \|\boldsymbol{\beta}_{j,-0}\|_{1,2} + \frac{1}{n} \Delta^\top \mathbf{W}^\top \boldsymbol{\epsilon}_j. \quad (\text{S22})$$

Recall that $V_l = \mathbf{w}_l^\top \boldsymbol{\epsilon}_j / n$, $l \in [(p-1)(q+1)]$, where $\mathbf{w}_l$ is the l th column of $\mathbf{W}$. Using similar arguments as in (S20) and again applying Lemma 4 gives that

$$\mathbb{P}\left(|V_l| > \frac{\lambda}{2}\right) \leq 2 \exp\left(-\frac{c\lambda^2 n}{4M^2 \phi_2 \sigma_{\epsilon_j}^2}\right) \leq 2 \exp\{-C''_0(\log p)\}, \quad (\text{S23})$$

where $C''_0 = cC^2/(4M^2 \phi_2)$, C is as defined in (12), and the last inequality is due to $\lambda^2 n \geq C^2 \sigma_{\epsilon_j}^2 \log p$. Applying the union bound inequality then gives

$$\mathbb{P}\left(\frac{1}{n} \|\mathbf{W}^\top \boldsymbol{\epsilon}_j\|_\infty \geq \frac{\lambda}{2}\right) \leq \mathbb{P}\left(\max_l |V_l| \geq \frac{\lambda}{2}\right) \leq 2 \exp\{\log q - (C''_0 - 1)(\log p)\}.$$

As $\log p \asymp \log q$, when C is chosen sufficiently large, we have, for some $c'_3 > 0$, $\exp\{\log q - (C''_0 - 1)(\log p)\} \leq \exp\{-c'_3 \log p\}$. Defining event $\mathcal{A}$ as

$$\mathcal{A} = \left\{ \frac{1}{n} \|\mathbf{W}^\top \boldsymbol{\epsilon}_j\|_\infty \leq \frac{\lambda}{2} \right\}, \quad (\text{S24})$$

we have $\mathbb{P}(\mathcal{A}^c) \leq 2 \exp\{-c'_3 \log p\}$. Given $\mathcal{A}$, (S22) leads to

$$2\|\hat{\boldsymbol{\beta}}_j\|_1 + 2\sqrt{\frac{s_j}{s_{j,g}}} \|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2} \leq 2\|\boldsymbol{\beta}_j\|_1 + 2\sqrt{\frac{s_j}{s_{j,g}}} \|\boldsymbol{\beta}_{j,-0}\|_{1,2} + \|\Delta\|_1,$$

as $\lambda_g = \sqrt{s_j/s_{j,g}}\lambda$ specified in (12).

Adding $\|\hat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}_j\|_1$ and $\sqrt{\frac{s_j}{s_{j,g}}} \|\hat{\boldsymbol{\beta}}_{j,-0} - \boldsymbol{\beta}_{j,-0}\|_{1,2}$ to both sides and noting $\|\Delta\|_1 = \|\hat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}_j\|_1$,

we obtain

$$\begin{aligned} & \|\hat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}_j\|_1 + 2\|\hat{\boldsymbol{\beta}}_j\|_1 + 2\sqrt{\frac{s_j}{s_{j,g}}} \|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2} + \sqrt{\frac{s_j}{s_{j,g}}} \|\hat{\boldsymbol{\beta}}_{j,-0} - \boldsymbol{\beta}_{j,-0}\|_{1,2} \\ & \leq 2\|\hat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}_j\|_1 + 2\|\boldsymbol{\beta}_j\|_1 + 2\sqrt{\frac{s_j}{s_{j,g}}} \|\boldsymbol{\beta}_{j,-0}\|_{1,2} + 2\sqrt{\frac{s_j}{s_{j,g}}} \|\hat{\boldsymbol{\beta}}_{j,-0} - \boldsymbol{\beta}_{j,-0}\|_{1,2}, \end{aligned}$$

which leads to

$$\begin{aligned} & \|\hat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}_j\|_1 + \sqrt{\frac{s_j}{s_{j,g}}} \|\hat{\boldsymbol{\beta}}_{j,-0} - \boldsymbol{\beta}_{j,-0}\|_{1,2} \\ & \leq 2(\|\hat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}_j\|_1 + \|\boldsymbol{\beta}_j\|_1 - \|\hat{\boldsymbol{\beta}}_j\|_1) + 2\sqrt{\frac{s_j}{s_{j,g}}} (\|\boldsymbol{\beta}_{j,-0} - \boldsymbol{\beta}_{j,-0}\|_{1,2} + \|\boldsymbol{\beta}_{j,-0}\|_{1,2} - \|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2}). \end{aligned}$$

Since $\|(\hat{\boldsymbol{\beta}}_j)_l - (\boldsymbol{\beta}_j)_l\|_1 + \|(\boldsymbol{\beta}_j)_l\|_1 - \|(\hat{\boldsymbol{\beta}}_j)_l\|_1 = 0$ for $l \in \mathcal{S}_j^c$ and $\|(\hat{\boldsymbol{\beta}}_j)_{(h)} - (\boldsymbol{\beta}_j)_{(h)}\|_{1,2} + \|(\boldsymbol{\beta}_j)_{(h)}\|_{1,2} - \|(\hat{\boldsymbol{\beta}}_j)_{(h)}\|_{1,2} = 0$ for $h \in \mathcal{G}_j^c$, using the triangular inequality yields that

$$\|\Delta_{\mathcal{S}_j^c}\|_1 \leq \|\Delta\|_1 \leq 4\|\Delta_{\mathcal{S}_j}\|_1 + 4\sqrt{\frac{s_j}{s_{j,g}}} \|\Delta_{\mathcal{G}_j}\|_{1,2}.$$

Therefore, conditional on event $\mathcal{A}$ in (S24), we have that

$$\|\Delta_{\mathcal{S}_j^c}\|_1 + \sqrt{\frac{s_j}{s_{j,g}}} \|\Delta_{\mathcal{G}_j^c}\|_{1,2} \leq 4\|\Delta_{\mathcal{S}_j}\|_1 + 4\sqrt{\frac{s_j}{s_{j,g}}} \|\Delta_{\mathcal{G}_j}\|_{1,2},$$

which further implies $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j \|\Delta_{\mathcal{S}_j}\|_1$ because $\|\Delta_{\mathcal{G}_j}\|_{1,2} \leq \|\Delta_{\mathcal{S}_j}\|_1$.

Step 2: The step bounds the diagonal and off-diagonal elements of Ψ . We first bound the diagonal elements, i.e., $\Psi_{ll} = \|\mathbf{w}_l\|_2^2/n$, where $w_l^{(i)}$ is sub-Gaussian as $u_l^{(i)}$ is bounded. Under Assumptions 1 and 2, we have $\text{Var}(w_l^{(i)}) \leq M^2\phi_2$ and by (S7), we have $\Sigma_{\mathbf{W}}(l, l) \geq \phi_1/\phi_0$. Using the concentration inequality for sub-exponential random variables in Lemma 4, we have

$$\mathbb{P}(|\Psi_{ll} - \Sigma_{\mathbf{W}}(l, l)| > \Sigma_{\mathbf{W}}(l, l)/2) \leq 2\exp(-c_4n),$$

for some positive constant c_4 . Immediately,

$$\mathbb{P}(\Psi_{ll} \notin [\phi_0/(2\phi_1), 2M^2\phi_2]) \leq 2\exp(-c_4n) \quad (\text{S25})$$

because

$$\mathbb{P}(\Psi_{ll} \notin [\phi_0/(2\phi_1), 2M^2\phi_2]) \leq \mathbb{P}(|\Psi_{ll} - \Sigma_{\mathbf{W}}(l, l)| > \Sigma_{\mathbf{W}}(l, l)/2).$$

Similarly, for the off diagonal elements, i.e., $\Psi_{kl} = \mathbf{w}_k^\top \mathbf{w}_l/n$, by noting that $\|\Sigma(\mathbf{u}^{(i)})\|_{\max} \leq \phi_2$, we have

$$\begin{aligned} & \mathbb{P}\left\{\Psi_{kl} \notin \left[-\frac{1}{c_0(1+8\eta_j)s_j}, \frac{3}{c_0(1+8\eta_j)s_j}\right]\right\} \\ & \leq \mathbb{P}(|\Psi_{kl} - \Sigma_{\mathbf{W}}(k, l)| \geq 2\Sigma_{\mathbf{W}}(k, l)) \leq 2\exp(-c_5n), \end{aligned} \quad (\text{S26})$$

for a positive constant c_5 .

Step 3: This step establishes that, conditional on event $\mathcal{A}$ and that

$$\Psi_{ll} \in [\phi_0/(2\phi_1), 2M^2\phi_2], \quad \Psi_{kl} \in \left[-\frac{1}{c_0(1+8\eta_j)s_j}, \frac{3}{c_0(1+8\eta_j)s_j}\right], \quad (\text{S27})$$

we have

$$\min_{\|\mathbf{v}_{\mathcal{S}_j^c}\|_1 \leq 3\eta_j \|\mathbf{v}_{\mathcal{S}_j}\|_1} \frac{\|\mathbf{W}\mathbf{v}\|_2}{\sqrt{n}\|\mathbf{v}_{\mathcal{S}_j}\|_2} \geq \sqrt{\frac{\phi_0}{2\phi_1} - \frac{1}{c_0}} > 0.$$

First, we have that

$$\begin{aligned} \frac{\|\mathbf{W}\mathbf{v}_{\mathcal{S}_j}\|_2^2}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} &= \frac{\mathbf{v}_{\mathcal{S}_j}^\top \text{diag}(\Psi)\mathbf{v}_{\mathcal{S}_j}}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} + \frac{\mathbf{v}_{\mathcal{S}_j}^\top (\Psi - \text{diag}(\Psi))\mathbf{v}_{\mathcal{S}_j}}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} \\ &\geq \frac{\phi_0}{2\phi_1} - \frac{1}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1^2}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2}, \end{aligned}$$

where $\text{diag}(\Psi)$ is a diagonal matrix with the diagonal elements being identical to those of Ψ , and the last inequality follows from (S27). Furthermore,

$$\begin{aligned} \frac{\|\mathbf{W}\mathbf{v}\|_2^2}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} &\geq \frac{\|\mathbf{W}\mathbf{v}_{\mathcal{S}_j}\|_2^2}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} + 2\frac{\mathbf{v}_{\mathcal{S}_j}^\top \Psi \mathbf{v}_{\mathcal{S}_j^c}}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2} \\ &\geq \frac{\phi_0}{2\phi_1} - \frac{1}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1^2}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} - \frac{2}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1 \|\mathbf{v}_{\mathcal{S}_j^c}\|_1}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} \\ &\geq \frac{\phi_0}{2\phi_1} - \frac{1+8\eta_j}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1^2}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} \geq \frac{\phi_0}{2\phi_1} - \frac{1}{c_0} > 0, \end{aligned}$$

where we have used the results that $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j\|\Delta_{\mathcal{S}_j}\|_1$ on event $\mathcal{A}$ and the fact that $\|\mathbf{v}_{\mathcal{S}_j}\|_1 \leq \sqrt{s_j}\|\mathbf{v}_{\mathcal{S}_j}\|_2$. Thus, we have finished Step 3.

Lastly, based on the results from Steps 1-3, we find the ℓ_∞ bound of the error of $\hat{\beta}_j$. For $l \in [(p-1)(q+1)]$, it is true that

$$\left(\Psi(\hat{\beta}_j - \beta_j)\right)_l = \Psi_{ll}(\hat{\beta}_j - \beta_j)_l + \sum_{k \neq l} \Psi_{kl}(\hat{\beta}_j - \beta_j)_k.$$

Given (S27), we have

$$\left| \left(\Psi(\hat{\beta}_j - \beta_j)\right)_l - \Psi_{ll}(\hat{\beta}_j - \beta_j)_l \right| \leq \frac{3}{c_0(1+8\eta_j)s_j} \sum_{k \neq l} |(\hat{\beta}_j - \beta_j)_k|,$$

and also

$$\|\hat{\beta}_j - \beta_j\|_\infty \leq 2\phi_1 \|\Psi\Delta\|_\infty + \frac{6\phi_1}{c_0\phi_0(1+8\eta_j)s_j} \|\Delta\|_1. \quad (\text{S28})$$

With $\Delta = \hat{\beta}_j - \beta_j$, and conditioning on $\|\Psi\Delta\|_\infty \leq \frac{3\eta_j\lambda}{2}$ and $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j\|\Delta_{\mathcal{S}_j}\|_1$ from Step 1, we have that

$$\frac{\|\mathbf{W}\Delta\|_2^2}{n} \leq \|\Psi\Delta\|_\infty \|\Delta\|_1 \leq \frac{3\eta_j\lambda}{2} (1+4\eta_j) \sqrt{s_j} \|\Delta_{\mathcal{S}_j}\|_2, \quad (\text{S29})$$

while Step 3 also gives that $\|\mathbf{W}\Delta\|_2^2/n \geq \{\phi_0/(2\phi_1) - 1/c_0\} \|\Delta_{\mathcal{S}_j}\|_2^2$. Combining the above two inequalities yields that $\{\phi_0/(2\phi_1) - 1/c_0\} \|\Delta_{\mathcal{S}_j}\|_2^2 \leq \frac{3\eta_j\lambda}{2} (1+4\eta_j) \sqrt{s_j} \|\Delta_{\mathcal{S}_j}\|_2$, and, therefore,

$$\|\Delta_{\mathcal{S}_j}\|_2 \leq 3\eta_j\lambda(1+4\eta_j) \frac{c_0\phi_1}{c_0\phi_0 - 2\phi_1} \sqrt{s_j}.$$

With $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j \|\Delta_{\mathcal{S}_j}\|_1$, it follows that $\|\Delta\|_1 \leq (1 + 4\eta_j) \|\Delta_{\mathcal{S}_j}\|_1 \leq (1 + 4\eta_j) \sqrt{s_j} \|\Delta_{\mathcal{S}_j}\|_2$, and, therefore,

$$\|\Delta\|_1 \leq 3\eta_j \lambda (1 + 4\eta_j)^2 \frac{c_0 \phi_1}{c_0 \phi_0 - 2\phi_1} s_j.$$

Plugging this into (S28), we get the desired result that with probability at least $1 - C'_1 \exp\{-C'_2 \log p\}$,

$$\|\hat{\beta}_j - \beta_j\|_\infty \leq \left(3\phi_1 \eta_j + \frac{18\phi_1^2 (1 + 4\eta_j)^2 \eta_j}{\phi_0 (c_0 \phi_0 - 2\phi_1) (1 + 8\eta_j)} \right) \lambda,$$

for some positive constants C'_1, C'_2 .

S3.3 Proof of Theorem 3

We start the proof by noting that the two events in (19) hold with the specified probability by applying Lemma 5, which is applicable because

$$\sup_{i \in [n]} \text{Var}(z_j^{(i)}) = \mathcal{O}(1), \quad \frac{1}{n} \sum_{i=1}^n \text{Var}(z_j^{(i)}) = \mathcal{O}(1),$$

due to $\{\Sigma(\mathbf{u}^{(i)})\}_{jj} \leq \phi_2, j \in [p]$ as assumed in Assumption 2. In what follows, we show (20), conditional on that the two events, as specified in (19), hold.

When Γ is unknown, compared to the oracle regression equation $\mathbf{z}_j = \mathbf{W}\beta_j + \epsilon_j$, we only have access to the noisy equation

$$\hat{\mathbf{z}}_j = \hat{\mathbf{W}}\beta_j + \mathbf{E}_j,$$

where $\mathbf{E}_j = \epsilon_j + (\hat{\mathbf{z}}_j - \mathbf{z}_j) + (\mathbf{W} - \hat{\mathbf{W}})\beta_j$. As $\hat{\beta}_j$ is a minimizer of the convex objective function (18), Lemma 1 implies

$$\begin{aligned} & \frac{1}{2n} \|\hat{\mathbf{z}}_j - \hat{\mathbf{W}}\hat{\beta}_j\|_2^2 + \lambda \|\hat{\beta}_j\|_1 + \lambda_g \|\hat{\beta}_{j,-0}\|_{1,2} + \frac{1}{2n} \|\hat{\mathbf{W}}(\hat{\beta}_j - \beta_j)\|_2^2 \\ & \leq \frac{1}{2n} \|\hat{\mathbf{z}}_j - \hat{\mathbf{W}}\beta_j\|_2^2 + \lambda \|\beta_j\|_1 + \lambda_g \|\beta_{j,-0}\|_{1,2}. \end{aligned}$$

With $\Delta = \hat{\beta}_j - \beta_j$, reorganizing terms in the above inequality leads to

$$\frac{1}{n} \|\hat{\mathbf{W}}\Delta\|_2^2 + \lambda \|\hat{\beta}_j\|_1 + \lambda_g \|\hat{\beta}_{j,-0}\|_{1,2} \leq \frac{1}{n} \langle \mathbf{E}_j, \hat{\mathbf{W}}\Delta \rangle + \lambda \|\beta_j\|_1 + \lambda_g \|\beta_{j,-0}\|_{1,2}.$$

Next, with $\Delta_{\mathbf{E}_j} = (\hat{\mathbf{z}}_j - \mathbf{z}_j) + (\mathbf{W} - \hat{\mathbf{W}})\boldsymbol{\beta}_j$, we have that

$$\begin{aligned} \frac{1}{n} \langle \mathbf{E}_j, \hat{\mathbf{W}} \Delta \rangle &= \frac{1}{n} \langle \boldsymbol{\epsilon}_j, \hat{\mathbf{W}} \Delta \rangle + \frac{1}{n} \langle \Delta_{\mathbf{E}_j}, \hat{\mathbf{W}} \Delta \rangle \\ &\leq \frac{1}{n} \langle \boldsymbol{\epsilon}_j, \hat{\mathbf{W}} \Delta \rangle + \frac{1}{2n} \|\Delta_{\mathbf{E}_j}\|_2^2 + \frac{1}{2n} \|\hat{\mathbf{W}} \Delta\|_2^2. \end{aligned}$$

Using similar arguments as in (S2), we obtain

$$\begin{aligned} &\frac{1}{2n} \|\hat{\mathbf{W}} \Delta\|_2^2 + \lambda \|\Delta_{\mathcal{S}_j^c}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j^c)}\|_{1,2} \\ &\leq \frac{1}{n} \langle \boldsymbol{\epsilon}_j, \hat{\mathbf{W}} \Delta \rangle + \frac{1}{2n} \|\Delta_{\mathbf{E}_j}\|_2^2 + \lambda \|\Delta_{\mathcal{S}_j}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j)}\|_{1,2}. \end{aligned} \quad (\text{S30})$$

Consider the two stochastic terms $\langle \boldsymbol{\epsilon}_j, \hat{\mathbf{W}} \Delta \rangle$ and $\|\Delta_{\mathbf{E}_j}\|_2^2$. For the latter, recall that our proof is conditional on the events of (19). Then, with probability at least $1 - \exp(\log p + \log q - \tau_1 \log q)$,

$$\frac{1}{n} \|\Delta_{\mathbf{E}_j}\|_2^2 \leq \frac{1}{n} \|\hat{\mathbf{z}}_j - \mathbf{z}_j\|_2^2 + \frac{\max_j \|\mathbf{W}_{\cdot j} - \hat{\mathbf{W}}_{\cdot j}\|_2^2}{n} \cdot \|\boldsymbol{\beta}_j\|_1^2 \lesssim \sigma_{\epsilon_j}^2 \frac{t \log q}{n}, \quad (\text{S31})$$

where the last inequality is true due to Assumption 5 and (19).

Next, we bound $\langle \boldsymbol{\epsilon}_j, \hat{\mathbf{W}} \Delta \rangle$. Defining $\hat{\mathcal{S}}_j = \left\{ l : (\hat{\boldsymbol{\beta}}_j)_l \neq 0, l \in [(p-1)(q+1)] \right\}$ and letting $\tilde{\mathcal{S}}_j = \mathcal{S}_j \cup \hat{\mathcal{S}}_j$, we have that

$$\begin{aligned} \langle \boldsymbol{\epsilon}_j, \hat{\mathbf{W}} \Delta \rangle &= \langle \boldsymbol{\epsilon}_j, \hat{\mathcal{P}}_{\tilde{\mathcal{S}}_j} \hat{\mathbf{W}}_{\tilde{\mathcal{S}}_j} \Delta_{\tilde{\mathcal{S}}_j} \rangle \\ &= \langle \hat{\mathcal{P}}_{\tilde{\mathcal{S}}_j} \boldsymbol{\epsilon}_j, \hat{\mathbf{W}} \Delta \rangle \leq \frac{1}{2a_1} \|\hat{\mathbf{W}} \Delta\|_2^2 + \frac{a_1}{2} \|\hat{\mathcal{P}}_{\tilde{\mathcal{S}}_j} \boldsymbol{\epsilon}_j\|_2^2, \end{aligned} \quad (\text{S32})$$

where $\hat{\mathcal{P}}_{\tilde{\mathcal{S}}_j}$ is the orthogonal projection matrix onto the column space of $\hat{\mathbf{W}}_{\tilde{\mathcal{S}}_j}$. Using the same argument as in (S4), we have

$$\|\hat{\mathcal{P}}_{\tilde{\mathcal{S}}_j} \boldsymbol{\epsilon}_j\|_2^2 < M \sigma_{\epsilon_j}^2 \{ (s_j + \hat{s}_j) \log(ep) + (s_{j,g} + \hat{s}_{j,g}) \log(eq/s_{j,g}) \} + \hat{r}, \quad (\text{S33})$$

where

$$\hat{r} = \sup_{\substack{1 \leq s \leq (p-1)(q+1) \\ 0 \leq s_g \leq q}} \left(\sup_{|\mathcal{J}|=s, |\mathcal{G}(\mathcal{J})|=s_g} \|\hat{\mathcal{P}}_{\mathcal{J}} \boldsymbol{\epsilon}_j\|_2^2 - M \sigma_{\epsilon_j}^2 \{ s \log(ep) + s_g \log(eq/s_g) \} \right)_+.$$

Setting $M = 9$, by Step 1 in Section S3.1, we have

$$\mathbb{P}\{\hat{r} \geq t\sigma_{\epsilon_j}^2\} < \sum_{s=1}^{(p-1)(q+1)} \sum_{s_g=0}^q c_1 \exp(-c_2 t) \exp[-3c_2 \{s \log(ep) + s_g \log(eq/s_g)\}].$$

We move to bound $\|\hat{\mathcal{P}}_{\hat{s}_j} \epsilon_j\|_2^2$ by using the computational optimality of $\hat{\beta}_j$. As in (S5) and (S6), it follows that

$$\lambda^2 \hat{s}_j + \lambda_g^2 \hat{s}_{j,g} \leq \frac{1}{n^2} \|\hat{\mathbf{W}}_{\hat{s}_j} (\hat{\mathbf{z}}_j - \hat{\mathbf{W}} \hat{\beta}_j)\|_2^2. \quad (\text{S34})$$

We also have that $\|\hat{\mathbf{W}}_{\hat{s}_j} - \mathbf{W}_{\hat{s}_j}\|/\sqrt{n} \lesssim \sqrt{\hat{s}_j t \log q/n}$ conditional on the events of (19). As $\hat{s}_j < s_\lambda = \mathcal{O}(n^{1/2})$, $t = o(n^{1/3})$ and $\log q = \mathcal{O}(n^{1/6})$, we have $\|\hat{\mathbf{W}}_{\hat{s}_j} - \mathbf{W}_{\hat{s}_j}\|/\sqrt{n} = o(1)$. Together with the result $\|\mathbf{W}_{\hat{s}_j}\|^2/n \leq M_1$ from Step 3 in Section S3.1, we have that $\|\hat{\mathbf{W}}_{\hat{s}_j}\|^2 \leq M_3$ for some $M_3 > 0$. It then follows from the Cauchy-Schwarz inequality that

$$\begin{aligned} \lambda^2 \hat{s}_j + \lambda_g^2 \hat{s}_{j,g} &\leq \frac{3M_3}{n} \|\hat{\mathbf{W}} \Delta\|_2^2 \\ &+ \frac{3M_3}{n} \|\hat{\mathcal{P}}_{\hat{s}_j} \epsilon_j\|_2^2 + \frac{3M_3}{n} \|\Delta_{\mathbf{E}_j}\|_2^2. \end{aligned} \quad (\text{S35})$$

Set $\lambda = C\sigma_{\epsilon_j} \sqrt{\{\log(ep)/n + s_{j,g} \log(eq/s_{j,g})/(ns_j)\}}$ and $\lambda_g = \sqrt{s_j/s_{j,g}} \lambda$, where $C = 3(a_2 M_3)^{1/2}$ for some $a_2 > 0$. Combining (S33) and (S35), we have that

$$\begin{aligned} (1 - \frac{3}{a_2}) \|\hat{\mathcal{P}}_{\hat{s}_j} \epsilon_j\|_2^2 &\leq 9\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\} \\ &+ \frac{3}{a_2} \|\hat{\mathbf{W}} \Delta\|_2^2 + \frac{3}{a_2} \|\Delta_{\mathbf{E}_j}\|_2^2 + \hat{r}. \end{aligned} \quad (\text{S36})$$

Plugging the above inequality and (S32) into (S30), we have

$$\begin{aligned} &\frac{\|\hat{\mathbf{W}} \Delta\|_2^2}{2n} + \lambda \|\Delta_{\mathcal{S}_j^c}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j^c)}\|_{1,2} \\ &\leq \frac{1}{2a_1} \frac{\|\hat{\mathbf{W}} \Delta\|_2^2}{n} + \frac{9a_1 a_2}{2(a_2 - 3)} \frac{\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}}{n} \\ &+ \frac{3a_1}{2(a_2 - 3)} \frac{\|\hat{\mathbf{W}} \Delta\|_2^2}{n} + \frac{a_1 a_2}{2(a_2 - 3)n} \hat{r} + \frac{1}{2n} \|\Delta_{\mathbf{E}_j}\|_2^2 \\ &+ \frac{a_1 a_2}{2(a_2 - 3)n} \|\Delta_{\mathbf{E}_j}\|_2^2 + \lambda \|\Delta_{\mathcal{S}_j}\|_1 + \lambda_g \|\Delta_{(\mathcal{G}_j)}\|_{1,2}. \end{aligned} \quad (\text{S37})$$

As in Section S3.1, we have that

$$\frac{\|\Delta_{\mathcal{S}_j}\|_1}{\sqrt{s_j}} + \frac{\|\Delta_{(\mathcal{G}_j)}\|_{1,2}}{\sqrt{s_{j,g}}} \leq \|\Delta_{\mathcal{S}_j}\|_2 + \|\Delta_{(\mathcal{G}_j)}\|_2 \leq 2\frac{\phi_1}{\phi_0}\|\Sigma_{\mathbf{W}}^{1/2}\Delta\|_2.$$

Consequently,

$$\lambda\|\Delta_{\mathcal{S}_j}\|_1 + \lambda_g\|\Delta_{(\mathcal{G}_j)}\|_{1,2} \leq 2C\frac{\phi_1}{\phi_0}\sqrt{e'_j}\|\Sigma_{\mathbf{W}}^{1/2}\Delta\|_2. \quad (\text{S38})$$

where $e'_j = \sigma_{\epsilon_j}^2\{s_j\log(ep) + s_{j,g}\log(eq/s_{j,g})\}/n$. Combining (S37) and (S38), we have

$$\begin{aligned} & \left\{ \frac{1}{2} - \frac{1}{2a_1} - \frac{3a_1}{2(a_2-3)} \right\} \frac{\|\hat{\mathbf{W}}\Delta\|_2^2}{n} \\ & \leq \left\{ \frac{9a_1a_2}{2(a_2-3)} + Ca_3\frac{\phi_1}{\phi_0} \right\} \frac{\sigma_{\epsilon_j}^2\{s_j\log(ep) + s_{j,g}\log(eq/s_{j,g})\}}{n} \\ & \quad + \frac{1}{a_3} \frac{\|\Sigma_{\mathbf{W}}^{1/2}\Delta\|_2^2}{n} + \frac{a_1a_2}{2(a_2-2)n}\hat{r} + C'' \left\{ \frac{1}{2n} + \frac{a_1a_2}{2(a_2-3)n} \right\}. \end{aligned}$$

Therefore, by choosing proper constants a_1 , a_2 and a_3 (e.g., $a_1 = 4$, $a_2 = 51$, $a_3 = 4$), we have, with probability at least $1 - c_1 \exp[-c'_2\{s_j\log(ep) + s_{j,g}\log(eq/s_{j,g})\}]$,

$$\frac{\|\hat{\mathbf{W}}\Delta\|_2^2}{n} - \frac{\|\Sigma_{\mathbf{W}}^{1/2}\Delta\|_2^2}{2n} \lesssim \frac{\sigma_{\epsilon_j}^2\{s_j\log(ep) + s_{j,g}\log(eq/s_{j,g})\}}{n}, \quad (\text{S39})$$

where we have used the fact that

$$\begin{aligned} & \mathbb{P} \left[\hat{r} \geq M_0\sigma_{\epsilon_j}^2\{s_j\log(ep) + s_{j,g}\log(eq/s_{j,g})\} \right] \\ & \leq c_1 \exp[-c'_2\{s_j\log(ep) + s_{j,g}\log(eq/s_{j,g})\}], \end{aligned}$$

for a large constant M_0 .

We then bound the difference between $\|\hat{\mathbf{W}}\Delta\|_2^2/n$ and $\|\Sigma_{\mathbf{W}}^{1/2}\Delta\|_2^2/n$. To do this, we first show that, with probability at least $1 - c_6 \exp[c_7\{\log p - (\tau_1 - 1)\log q\}]$,

$$\sup_{\mathbf{v} \in \mathbb{K}_0(2C_{\beta_j}s_j)} \left| \mathbf{v}^\top \left(\frac{\hat{\mathbf{W}}^\top \hat{\mathbf{W}}}{n} - \Sigma_{\mathbf{W}} \right) \mathbf{v} \right| \leq 1/L, \quad (\text{S40})$$

where L is a large constant and $\mathbb{K}_0(2C_{\beta_j}s_j) = \{\mathbf{v} : \|\mathbf{v}\|_0 \leq 2C_{\beta_j}s_j \text{ and } \|\mathbf{v}\|_2 = 1\}$ for some positive constant C_{β_j} . Notice that

$$\left| \mathbf{v}^\top \left(\frac{\hat{\mathbf{W}}^\top \hat{\mathbf{W}}}{n} - \frac{\mathbf{W}^\top \mathbf{W}}{n} \right) \mathbf{v} \right| \leq \frac{2|\mathbf{v}^\top \mathbf{W}^\top (\hat{\mathbf{W}} - \mathbf{W}) \mathbf{v}|}{n} + \frac{\|(\hat{\mathbf{W}} - \mathbf{W})\mathbf{v}\|_2^2}{n},$$

where we have used that $\|\mathbf{v}\|_1 \leq \sqrt{2C_{\beta_j} s_j} \|\mathbf{v}\|_2$. For the second term on the right hand-side, we have with probability at least $1 - 3 \exp\{\log p - (\tau_1 - 1) \log q\}$,

$$\frac{\|(\hat{\mathbf{W}} - \mathbf{W})\mathbf{v}\|_2}{\sqrt{n}} \leq \frac{\max_j \|\mathbf{W}_{\cdot j} - \hat{\mathbf{W}}_{\cdot j}\|_2}{\sqrt{n}} \cdot \|\mathbf{v}\|_1 \lesssim \sqrt{\frac{2C_{\beta_j} s_j t \log q}{n}} = o(1),$$

and, moreover,

$$\frac{|\mathbf{v}^\top \mathbf{W}^\top (\hat{\mathbf{W}} - \mathbf{W}) \mathbf{v}|}{n} \leq \frac{\|(\hat{\mathbf{W}} - \mathbf{W})\mathbf{v}\|_2}{\sqrt{n}} \times \frac{\|\mathbf{W}\mathbf{v}\|_2}{\sqrt{n}}. \quad (\text{S41})$$

Next, using a similar argument as in Section S3.1, it follows from (S8) and Lemma 6 that with probability at least $1 - \mathcal{O}\{(pq)^{-1}\}$

$$\left| \mathbf{v}^\top \left(\frac{\mathbf{W}^\top \mathbf{W}}{n} - \boldsymbol{\Sigma}_{\mathbf{W}} \right) \mathbf{v} \right| = o(1).$$

Using the same argument as in (S7) and by Assumption 5, we can show $\|\boldsymbol{\Sigma}_{\mathbf{W}}\|$ is upper bounded by $\phi'_0 \phi_1$. Thus, we have that $\|\mathbf{W}\mathbf{v}\|_2 / \sqrt{n} = \mathcal{O}(1)$ with probability at least $1 - \mathcal{O}\{(pq)^{-1}\}$. Putting this together with (S41), we have shown (S40).

Next, conditioning on (S40) and using the result in Lemma 7, we have

$$\left| \Delta^\top \left(\frac{\hat{\mathbf{W}}^\top \hat{\mathbf{W}}}{n} - \boldsymbol{\Sigma}_{\mathbf{W}} \right) \Delta \right| \leq \frac{1}{L'} \left(\|\Delta\|_2^2 + \frac{1}{s_j} \|\Delta\|_1^2 \right). \quad (\text{S42})$$

Plugging this into (S39), we have

$$\begin{aligned} \frac{\|\mathbf{W}\Delta\|_2^2}{2n} &\lesssim \frac{\sigma_{\epsilon_j}^2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}}{n} \\ &+ \frac{1}{L'} \left(\|\Delta\|_2^2 + \frac{1}{s_j} \|\Delta\|_1^2 \right) + \frac{\sigma_{\epsilon_j}^2}{n}. \end{aligned} \quad (\text{S43})$$

By choosing an appropriate a_1 given a_2 in (S37), we have with probability at least $1 - c_1 \exp[-c'_2 \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\}]$,

$$\frac{\|\Delta_{\mathcal{S}_j^c}\|_1}{\sqrt{s_j}} + \frac{\|\Delta_{(\mathcal{G}_j^c)}\|_{1,2}}{\sqrt{s_{j,g}}} \leq \sqrt{e'_j} + \frac{\|\Delta_{\mathcal{S}_j}\|_1}{\sqrt{s_j}} + \frac{\|\Delta_{(\mathcal{G}_j)}\|_{1,2}}{\sqrt{s_{j,g}}}. \quad (\text{S44})$$

Adding $\|\Delta_{\mathcal{S}_j}\|_1 / \sqrt{s_j}$ to both sides of (S44), we get

$$\frac{\|\Delta\|_1}{\sqrt{s_j}} \leq \sqrt{e'_j} + 3\|\Delta\|_2 \quad (\text{S45})$$

Plugging (S45) into (S43) and by (S7), we have

$$\|\Delta\|_2^2 \lesssim \frac{\sigma_{\epsilon_j}^2}{n} \{s_j \log(ep) + s_{j,g} \log(eq/s_{j,g})\} + \frac{\sigma_{\epsilon_j}^2}{n},$$

with probability at least $1 - C_3 \exp[C_4\{\log p - (\tau_1 - 1)\log q\}]$, for some positive constants C_3 and C_4 .

□

S3.4 Proof of Theorem 4

We first establish the ℓ_∞ norm bound of $\hat{\beta}_j$ in (21) with three steps.

Step 1: In this step, we show that, with probability at least $1 - c_8 \exp[c_9\{\log p - (\tau_1 - 1)\log q\}]$ for some $c_8, c_9 > 0$,

$$\|\hat{\Psi}\Delta\|_\infty \leq \frac{3\eta_j\lambda}{2}. \quad (\text{S46})$$

And it also holds with probability at least $1 - c'_8 \exp[c'_9\{\log p - (\tau_1 - 1)\log q\}]$ that, for some $c'_8, c'_9 > 0$,

$$\|\Delta_{S_j^c}\|_1 \leq 4\eta_j \|\Delta_{S_j}\|_1. \quad (\text{S47})$$

Under the KKT conditions, if θ is an optimum of (18), then

$$\begin{cases} \left(\hat{\mathbf{W}}^\top(\hat{\mathbf{z}}_j - \hat{\mathbf{W}}\theta)/n\right)_l = \text{sign}(\theta_l)\lambda & \text{if } \theta_l \neq 0, l \in (0) \\ \left(\hat{\mathbf{W}}^\top(\hat{\mathbf{z}}_j - \hat{\mathbf{W}}\theta)/n\right)_l = \text{sign}(\theta_l)\lambda + \lambda_g \frac{\theta_l}{\|\theta_{(h)}\|_2} & \text{if } \theta_l \neq 0, l \in (h) \\ \left|\left(\hat{\mathbf{W}}^\top(\hat{\mathbf{z}}_j - \hat{\mathbf{W}}\theta)/n\right)_l\right| < \eta_j\lambda & \text{if } \theta_l = 0. \end{cases}$$

As such, any solution $\hat{\beta}_j$ satisfies that

$$\left\| \frac{1}{n} \hat{\mathbf{W}}^\top (\hat{\mathbf{z}}_j - \hat{\mathbf{W}}\hat{\beta}_j) \right\|_\infty \leq \eta_j\lambda.$$

If we can show that with high probability

$$\frac{1}{n} \left\| \hat{\mathbf{W}}^\top \mathbf{E}_j \right\|_\infty \leq \frac{\eta_j\lambda}{2}, \quad (\text{S48})$$

then we can reach the desired conclusion that with high probability

$$\left\| \hat{\Psi} \Delta \right\|_{\infty} \leq \frac{3\eta_j \lambda}{2}.$$

Now we consider the inequality in (S48), and note that

$$\frac{1}{n} \left\| \hat{\mathbf{W}}^{\top} \mathbf{E}_j \right\|_{\infty} \leq \underbrace{\frac{1}{n} \left\| \mathbf{W}^{\top} \boldsymbol{\epsilon}_j \right\|_{\infty}}_{\text{I}} + \underbrace{\frac{1}{n} \left\| (\hat{\mathbf{W}} - \mathbf{W})^{\top} \boldsymbol{\epsilon}_j \right\|_{\infty}}_{\text{II}} + \underbrace{\frac{1}{n} \left\| \hat{\mathbf{W}}^{\top} \Delta_{\mathbf{E}_j} \right\|_{\infty}}_{\text{III}}.$$

Consider term (I). Define $V_l = \mathbf{w}_l^{\top} \boldsymbol{\epsilon}_j / n$, $j \in [(p-1)(q+1)]$. Using a similar argument as in (S21), we have

$$\mathbb{P} \left(|V_l| > \frac{\eta_j \lambda}{2} \right) \leq 2 \exp \left(-\frac{c\eta_j^2 \lambda^2 n}{4M^2 \phi_2 \sigma_{\epsilon_j}^2} \right) \leq 2 \exp \{ -C'_0 (\log p + \log q) \},$$

where $C'_0 = cC/(4M^2 \phi_2) > 1$. Using the union bound inequality, we have

$$\begin{aligned} \mathbb{P} \left(\frac{1}{n} \left\| \mathbf{W}^{\top} \boldsymbol{\epsilon}_j \right\|_{\infty} \geq \frac{\eta_j \lambda}{2} \right) &\leq \mathbb{P} \left(\max_l |V_l| \geq \frac{\eta_j \lambda}{2} \right) \\ &\leq 2 \exp \{ -(C'_0 - 1) (\log p + \log q) \}. \end{aligned}$$

For term (II), it is true that

$$\frac{1}{n} \left\| (\hat{\mathbf{W}} - \mathbf{W})^{\top} \boldsymbol{\epsilon}_j \right\|_{\infty} = \frac{1}{n} \max_l |\langle \hat{\mathbf{w}}_l - \mathbf{w}_l, \boldsymbol{\epsilon}_j \rangle| \leq \frac{\max_l \|\hat{\mathbf{w}}_l - \mathbf{w}_l\|_2}{\sqrt{n}} \cdot \frac{\|\boldsymbol{\epsilon}_j\|_2}{\sqrt{n}}.$$

Using Lemma 4, we have that for any large constant $M_4 > 0$,

$$\mathbb{P} \left(\frac{\|\boldsymbol{\epsilon}_j\|_2}{\sqrt{n}} > M_4^{1/2} \sigma_{\epsilon_j} \right) \leq \mathbb{P} \left(\left| \frac{\|\boldsymbol{\epsilon}_j\|_2^2}{n} - \sigma_{\epsilon_j}^2 \right| > (M_4 - 1) \sigma_{\epsilon_j}^2 \right) \leq 2 \exp(-b_0 \log q).$$

We further have $\mathbb{P}(\|\hat{\mathbf{w}}_l - \mathbf{w}_l\|_2 / \sqrt{n}) \lesssim t^{1/2} \lambda_1$ with probability at least $1 - 3 \exp(-\tau_1 \log q)$

and $t^{1/2} \lambda_1 = o(\eta_j \lambda)$, which is true as $t = o(\sqrt{n/\log q})$. Applying the union bound, we have

$$\mathbb{P} \left(\frac{1}{n} \left\| (\hat{\mathbf{W}} - \mathbf{W})^{\top} \boldsymbol{\epsilon}_j \right\|_{\infty} \geq \frac{\eta_j \lambda}{2} \right) \leq b'_0 \exp \{ \log p + \log q - \tau_1 \log q \}. \quad (\text{S49})$$

Moving to term (III), notice that there exists a large constant $M'_4 > 0$ such that

$$\begin{aligned} \frac{1}{n} \left\| \hat{\mathbf{W}}^{\top} \Delta_{\mathbf{E}_j} \right\|_{\infty} &= \left\| (\mathbf{W} + \hat{\mathbf{W}} - \mathbf{W})^{\top} \Delta_{\mathbf{E}_j} / n \right\|_{\infty} \\ &\leq \frac{1}{n} \left\| \mathbf{W}^{\top} \Delta_{\mathbf{E}_j} \right\|_{\infty} + \frac{\max_l \|\hat{\mathbf{w}}_l - \mathbf{w}_l\|_2}{\sqrt{n}} \cdot \frac{\|\Delta_{\mathbf{E}_j}\|_2}{\sqrt{n}} \\ &\leq \frac{1}{n} \left\| \mathbf{W}^{\top} \Delta_{\mathbf{E}_j} \right\|_{\infty} + M'_4 t \lambda_1^2, \end{aligned}$$

with probability at least $b_0'' \exp\{\log p + \log q - \tau_1 \log q\}$, where the last inequality follows from (19) and (S31). Since $t^{1/2} \lambda_1 = o(\eta_j \lambda)$, it suffices to bound the term of $\|\mathbf{W}^\top \Delta_{\mathbf{E}_j}\|/n$. To this end, we have

$$\frac{1}{n} \|\mathbf{W}^\top \Delta_{\mathbf{E}_j}\|_\infty < \frac{\max_l \|\mathbf{w}_l\|_2}{\sqrt{n}} \cdot \frac{\|\Delta_{\mathbf{E}_j}\|_2}{\sqrt{n}}.$$

As $\mathbf{w}_l$ has independent sub-Gaussian entries with a bounded sub-Gaussian norm, using a similar argument as for term (II) we have

$$\mathbb{P} \left(\frac{1}{n} \|\mathbf{W}^\top \Delta_{\mathbf{E}_j}\|_\infty \geq \frac{\eta_j \lambda}{2} \right) \leq b_0''' \exp\{\log p + \log q - \tau_1 \log q\}. \quad (\text{S50})$$

Combining terms (I)-(III), we have finished showing (S46) of Step 1.

Next, we prove that $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j \|\Delta_{\mathcal{S}_j}\|_1$. By the definition of $\hat{\boldsymbol{\beta}}_j$, we have

$$\frac{1}{2n} \|\hat{\mathbf{z}}_j - \hat{\mathbf{W}} \hat{\boldsymbol{\beta}}_j\|_2^2 + \lambda \|\hat{\boldsymbol{\beta}}_j\|_1 + \lambda_g \|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2} \leq \frac{1}{2n} \|\mathbf{E}_j\|_2^2 + \lambda \|\boldsymbol{\beta}_j\|_1 + \lambda_g \|\boldsymbol{\beta}_{j,-0}\|_{1,2}.$$

Developing the left hand side of the above inequality, we have

$$\lambda \|\hat{\boldsymbol{\beta}}_j\|_1 + \lambda_g \|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2} \leq \lambda \|\boldsymbol{\beta}_j\|_1 + \lambda_g \|\boldsymbol{\beta}_{j,-0}\|_{1,2} + \frac{1}{n} \Delta^\top \hat{\mathbf{W}}^\top \mathbf{E}_j.$$

Define an event

$$\mathcal{A}_1 = \left\{ \frac{1}{n} \|\mathbf{W}^\top \Delta_{\mathbf{E}_j}\|_\infty \leq \frac{\lambda}{2} \right\}. \quad (\text{S51})$$

We again write

$$\frac{1}{n} \|\hat{\mathbf{W}}^\top \mathbf{E}_j\|_\infty \leq \underbrace{\frac{1}{n} \|\mathbf{W}^\top \boldsymbol{\epsilon}_j\|_\infty}_{\text{I}} + \underbrace{\frac{1}{n} \|(\hat{\mathbf{W}} - \mathbf{W})^\top \boldsymbol{\epsilon}_j\|_\infty}_{\text{II}} + \underbrace{\frac{1}{n} \|\hat{\mathbf{W}}^\top \Delta_{\mathbf{E}_j}\|_\infty}_{\text{III}}.$$

For term (I), first define $V_l = \mathbf{w}_l^\top \boldsymbol{\epsilon}_j / n$, $j \in [(p-1)(q+1)]$. Using a similar argument as in (S23), we have

$$\mathbb{P} \left(\frac{1}{n} \|\mathbf{W}^\top \boldsymbol{\epsilon}_j\|_\infty \geq \frac{\lambda}{2} \right) \leq \mathbb{P} \left(\max_l |V_l| \geq \frac{\lambda}{2} \right) \leq \exp\{-c'_3 \log p\};$$

for term (II), using the same argument as in (S49) and by noting $t^{1/2}\lambda_1 = o(\lambda)$, which is true as $t = o(\sqrt{n/\log q})$, we have

$$\mathbb{P}\left(\frac{1}{n}\|(\hat{\mathbf{W}} - \mathbf{W})^\top \boldsymbol{\epsilon}_j\|_\infty \geq \frac{\lambda}{2}\right) \leq b'_0 \exp\{\log p + \log q - \tau_1 \log q\};$$

for term (III), using the same argument as in (S50) and again noting $t^{1/2}\lambda_1 = o(\lambda)$, we have

$$\mathbb{P}\left(\frac{1}{n}\|\mathbf{W}^\top \Delta_{\mathbf{E}_j}\|_\infty \geq \frac{\lambda}{2}\right) \leq b''_0 \exp\{\log p + \log q - \tau_1 \log q\}.$$

Combining terms (I)-(III), we have $\mathbb{P}(\mathcal{A}_1^c) \leq c'_8 \exp[c'_9\{\log p - (\tau_1 - 1)\log q\}]$. Conditioning on $\mathcal{A}_1$ in (S51), we have that

$$2\|\hat{\boldsymbol{\beta}}_j\|_1 + 2\sqrt{\frac{s_j}{s_{j,g}}}\|\hat{\boldsymbol{\beta}}_{j,-0}\|_{1,2} \leq 2\|\boldsymbol{\beta}_j\|_1 + 2\sqrt{\frac{s_j}{s_{j,g}}}\|\boldsymbol{\beta}_{j,-0}\|_{1,2} + \|\Delta\|_1.$$

The rest of the arguments is similar to Step 1 in the proof of Theorem 2, which leads us to the desired result of $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j\|\Delta_{\mathcal{S}_j}\|_1$ given $\mathcal{A}_1$.

Step 2: In this step, we bound the diagonal and off-diagonal elements of $\hat{\boldsymbol{\Psi}}$. First, we consider the diagonal elements of $\hat{\boldsymbol{\Psi}}$, i.e., $\hat{\boldsymbol{\Psi}}_{ll}$'s. Note that

$$\frac{\|\mathbf{w}_l\|_2}{\sqrt{n}} - \frac{\|\hat{\mathbf{w}}_l - \mathbf{w}_l\|_2}{\sqrt{n}} \leq \frac{\|\hat{\mathbf{w}}_l\|_2}{\sqrt{n}} \leq \frac{\|\mathbf{w}_l\|_2}{\sqrt{n}} + \frac{\|\hat{\mathbf{w}}_l - \mathbf{w}_l\|_2}{\sqrt{n}}.$$

Since $\|\hat{\mathbf{w}}_l - \mathbf{w}_l\|_2/\sqrt{n} = \mathcal{O}(t^{1/2}\lambda_1) = o(1)$ and $\mathbb{P}(\boldsymbol{\Psi}_l \notin [\phi_0/(2\phi_1), 2M^2\phi_2]) \leq 2\exp(-c_4n)$ from (S25), we have that $\phi_0/(3\phi_1) \leq \hat{\boldsymbol{\Psi}}_{ll} \leq 3M^2\phi_2$ with probability at least $1 - 2\exp(-c'_4n)$.

Next, we consider the off-diagonal elements. It holds for $k \neq l$ that

$$\|\mathbf{w}_k^\top \mathbf{w}_l - \hat{\mathbf{w}}_k^\top \hat{\mathbf{w}}_l\|_2 \leq \|\mathbf{w}_k^\top (\mathbf{w}_l - \hat{\mathbf{w}}_l)\|_2 + \|\hat{\mathbf{w}}_l^\top (\hat{\mathbf{w}}_k - \mathbf{w}_k)\|_2.$$

Since $\|\hat{\mathbf{w}}_l - \mathbf{w}_l\|_2/\sqrt{n} = \mathcal{O}(t^{1/2}\lambda_1) = o(1)$ and (S26) shows

$$\mathbb{P}\left\{\boldsymbol{\Psi}_{kl} \notin \left[-\frac{1}{c_0(1 + 8\eta_j)s_j}, \frac{3}{c_0(1 + 8\eta_j)s_j}\right]\right\} \leq 2\exp(-c_5n),$$

the following holds with probability at least $1 - \exp(-c'_5 n)$,

$$\max_{k \neq l} \hat{\Psi}_{kl} \in \left[-\frac{2}{c_0(1+8\eta_j)s_j}, \frac{4}{c_0(1+8\eta_j)s_j}\right].$$

Step 3: We show that conditional on $\mathcal{A}_1$ and that

$$\max_l \hat{\Psi}_{ll} \in [\phi_0/(3\phi_1), 3M^2\phi_2], \quad \max_{k \neq l} \hat{\Psi}_{kl} \in \left[-\frac{2}{c_0(1+8\eta_j)s_j}, \frac{4}{c_0(1+8\eta_j)s_j}\right], \quad (\text{S52})$$

it holds that

$$\min_{\|\mathbf{v}_{\mathcal{S}_j^c}\|_1 \leq 3\eta_j \|\mathbf{v}_{\mathcal{S}_j}\|_1} \frac{\|\hat{\mathbf{W}}\mathbf{v}\|_2}{\sqrt{n}\|\mathbf{v}_{\mathcal{S}_j}\|_2} \geq \sqrt{\frac{\phi_0}{3\phi_1} - \frac{2}{c_0}} > 0.$$

First, given (S52), we have that

$$\begin{aligned} \frac{\|\hat{\mathbf{W}}\mathbf{v}_{\mathcal{S}_j}\|_2^2}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} &= \frac{\mathbf{v}_{\mathcal{S}_j}^\top \text{diag}(\hat{\Psi})\mathbf{v}_{\mathcal{S}_j}}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} + \frac{\mathbf{v}_{\mathcal{S}_j}^\top (\hat{\Psi} - \text{diag}(\hat{\Psi}))\mathbf{v}_{\mathcal{S}_j}}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} \\ &\geq \frac{\phi_0}{3\phi_1} - \frac{2}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1^2}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2}. \end{aligned}$$

Furthermore, given (S52) and $\mathcal{A}_1$, we have that

$$\begin{aligned} \frac{\|\hat{\mathbf{W}}\mathbf{v}\|_2^2}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} &\geq \frac{\|\hat{\mathbf{W}}\mathbf{v}_{\mathcal{S}_j}\|_2^2}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} + 2 \frac{\mathbf{v}_{\mathcal{S}_j}^\top \hat{\Psi} \mathbf{v}_{\mathcal{S}_j^c}}{n\|\mathbf{v}_{\mathcal{S}_j}\|_2} \\ &\geq \frac{\phi_0}{3\phi_1} - \frac{2}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1^2}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} - \frac{4}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1 \|\mathbf{v}_{\mathcal{S}_j^c}\|_1}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} \\ &\geq \frac{\phi_0}{3\phi_1} - \frac{2(1+8\eta_j)}{c_0(1+8\eta_j)s_j} \frac{\|\mathbf{v}_{\mathcal{S}_j}\|_1^2}{\|\mathbf{v}_{\mathcal{S}_j}\|_2^2} \geq \frac{\phi_0}{3\phi_1} - \frac{2}{c_0} > 0, \end{aligned}$$

where we have used the results that $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j \|\Delta_{\mathcal{S}_j}\|_1$ and the fact that $\|\mathbf{v}_{\mathcal{S}_j}\|_1 \leq \sqrt{s_j} \|\mathbf{v}_{\mathcal{S}_j}\|_2$.

Lastly, with results from Steps 1-3, we find the ℓ_∞ bound of β_j . For $l \in [(p-1)(q+1)]$, it is true that

$$\left(\hat{\Psi}(\hat{\beta}_j - \beta_j)\right)_l = \hat{\Psi}_{ll}(\hat{\beta}_j - \beta_j)_l + \sum_{k \neq l} \hat{\Psi}_{kl}(\hat{\beta}_j - \beta_j)_k$$

Given (S52) from Step 2, we have

$$\left|\left(\hat{\Psi}(\hat{\beta}_j - \beta_j)\right)_l - \hat{\Psi}_{ll}(\hat{\beta}_j - \beta_j)_l\right| \leq \frac{4}{c_0(1+8\eta_j)s_j} \sum_{k \neq l} |(\hat{\beta}_j - \beta_j)_k|,$$

SNP	co-expressed genes
rs10492975	(CALML5,PIK3R2), (CALML5,CAMK1)
rs723211	(HRAS,CALML5), (CALML5,CAMK1G)
rs1347069	(SHC4,CDKN2A)
rs473698	(PRKCG,CAMK1)
rs4118334	(SHC2,CAMK1)
rs882664	(PRKCA,CAMK1)
rs1267622	(SHC3,RAF1)

Table S6: Identified co-expression QTLs and the corresponding co-expressed genes.

and also

$$\|\hat{\beta}_j - \beta_j\|_\infty \leq 3\phi_1 \left\| \hat{\Psi} \Delta \right\|_\infty + \frac{12\phi_1}{c_0\phi_0(1+8\eta_j)s_j} \|\Delta\|_1. \quad (\text{S53})$$

With $\Delta = \hat{\beta}_j - \beta_j$ and given $\left\| \hat{\Psi} \Delta \right\|_\infty \leq \frac{3\eta_j\lambda}{2}$ and $\|\Delta_{\mathcal{S}_j^c}\|_1 \leq 4\eta_j\|\Delta_{\mathcal{S}_j}\|_1$ from Step 1, we have

$$\frac{\|\hat{\mathbf{W}}\Delta\|_2^2}{n} \leq \|\hat{\Psi}\Delta\|_\infty\|\Delta\|_1 \leq \frac{3\eta_j\lambda}{2}(4\eta_j+1)\sqrt{s_j}\|\Delta_{\mathcal{S}_j}\|_1.$$

We also have from Step 3 that $\|\hat{\mathbf{W}}\Delta\|_2^2/n \geq \{\phi_0/(3\phi_1) - 2/c_0\}\|\Delta_{\mathcal{S}_j}\|_2^2$ given (S52) and $\mathcal{A}_1$.

Combining these two inequalities and by noting $\|\Delta\|_1 \leq (1+4\eta_j)\sqrt{s_j}\|\Delta_{\mathcal{S}_j}\|_2$, we have that

$$\|\Delta\|_1 \leq \frac{3\eta_j\lambda}{2}(1+4\eta_j)^2 \frac{3c_0\phi_1}{c_0\phi_0 - 6\phi_1} s_j.$$

Plugging this into (S53), we obtain that

$$\|\hat{\beta}_j - \beta_j\|_\infty \leq \frac{9}{2} \left\{ \phi_1\eta_j + \frac{12\phi_1^2(1+4\eta_j)^2}{\phi_0(c_0\phi_0 - 6\phi_1)(1+8\eta_j)} \right\} \lambda.$$

with probability at least $1 - C_5 \exp[C_6\{\log p - (\tau_1 - 1)\log q\}]$, for some positive constants C_5 and C_6 .

S4 Additional results of data analysis

Additional references

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